Four fundamental interactions:

1. Gravitation
2. Electromagnetic
3. Weak
4. Strong

(3–4: nuclear)

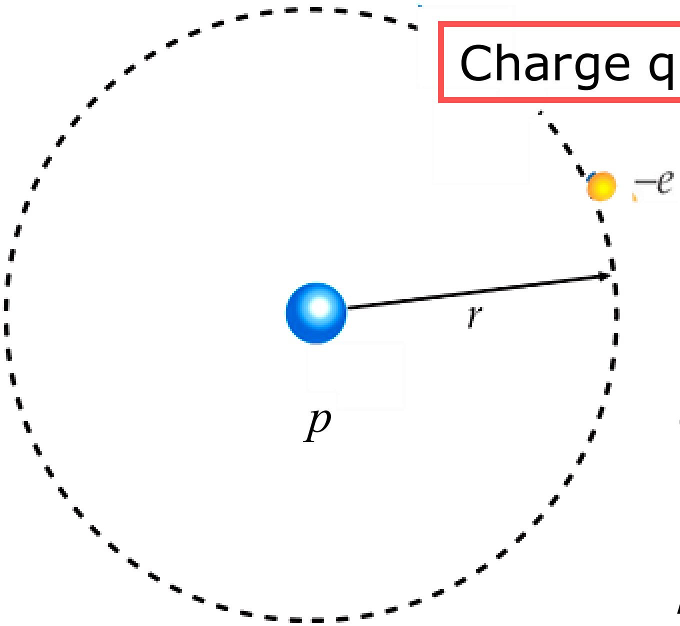
Electromagnetic interactions

Electric field — Magnetic field

$$\underbrace{\vec{\mathbf{E}}\ ,\ Q \qquad \qquad \vec{\mathbf{B}},\ \cancel{Q_M}}_{\text{Maxwell equations}}$$

ELECTROSTATIC INTERACTIONS

Charge quantization



Each electron has a mass = m_e and charge = $-e$

Each proton has a mass = m_p and charge = e

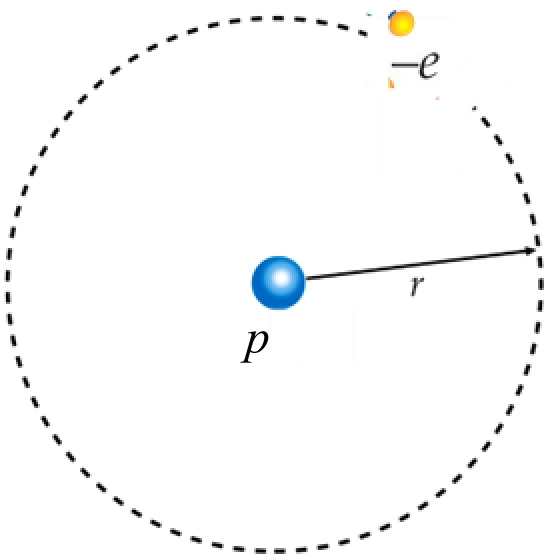
In Benjamin Franklin's days, the electric charge was thought to be a continuous fluid – an idea that was useful for many purposes. Experiments show that the „electrical fluid" is made up of multiples of a certain elementary charge.

Any positive or negative charge Q can be written as:

$Q=Ne$, $N=\ldots, -3, -2, -1, +1, +2, +3$

Elementary charge $e=1.602\cdot10^{-19}$ C

Conservation of Electric Charge



Total charge of an isolated system, i.e., algebraic sum of positive and negative electric charges, is constant in time

$$\boxed{Q_{tot}=\text{const}}$$

$$Q_{całk} = Q_e + Q_p$$

$$= -e + e$$

$$= 0$$

Examples of charge conservation

- Radioactive decay of a nucleus

$$ {}^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} $$

Atomic number Z=92 means 92 protons in the nucleus and its charge 92e

Emission of α particle

Charge conservation 92e=90e+2e

Examples of charge conservation

- Annihilation of e$^-$ and its antiparticle - positron e+

$$e^- + e^+ \rightarrow \gamma + \gamma$$

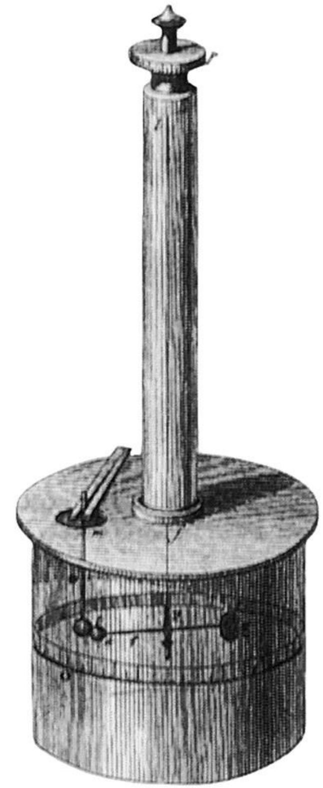
Emission of two photons

- Creation of a pair

$$\gamma \rightarrow e^- + e^+$$

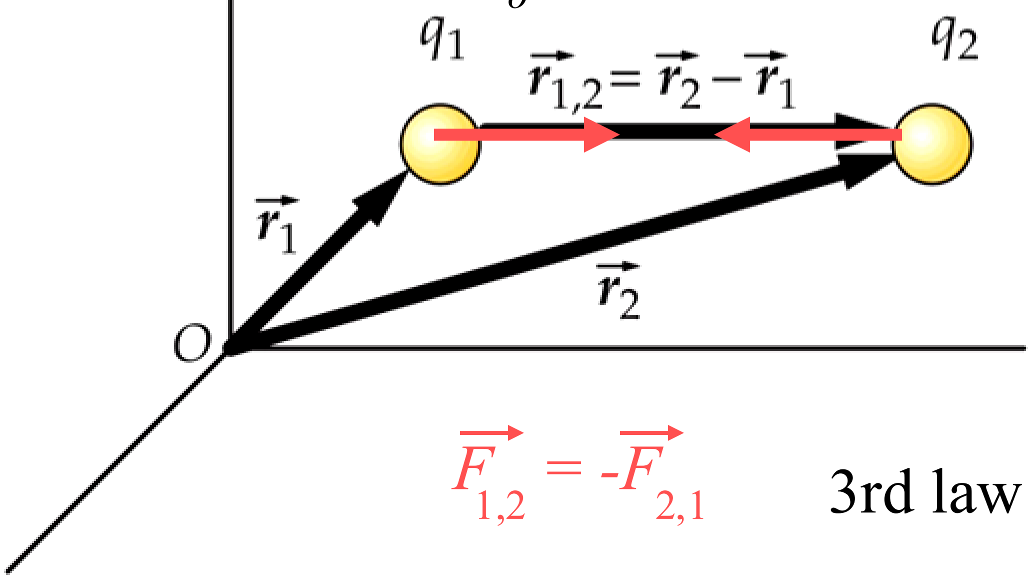
Empirical Coulomb's law

$$\vec{\mathbf{F}}_{1,2} = \frac{k\, q_1 q_2}{r_{1,2}^2}\,\hat{\mathbf{r}}_{1,2} = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r_{1,2}^2}\,\hat{\mathbf{r}}_{1,2}$$



1785 – instrument

$$\mathrm{k} = \frac{1}{4\pi\varepsilon_o} \text{ where } \varepsilon_o = 8.85\cdot 10^{-12}\,\frac{\mathrm{C}^2}{\mathrm{Nm}^2}$$



Charles Coulomb 1736-1806

$$\overrightarrow{F}_{1,2} = -\overrightarrow{F}_{2,1}$$

3rd law of dynamics

ELECTROSTATIC AND GRAVITATIONAL FIELD

SIMILARITIES

$$\vec{\mathbf{F}} = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}$$

$$\vec{\mathbf{F}} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$$

COULOMB'S LAW

NEWTON'S LAW

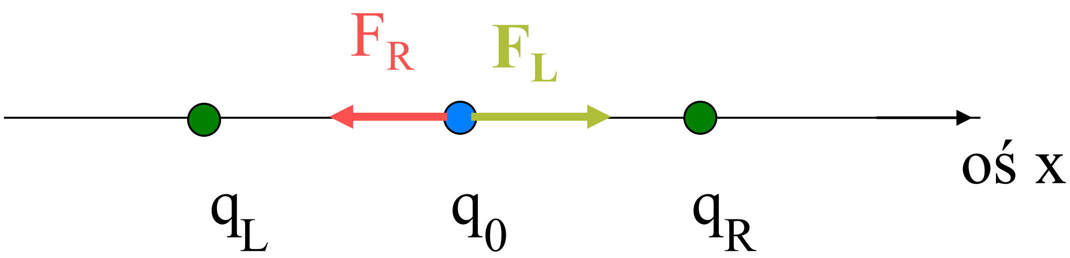
$$k = \frac{1}{4\pi\varepsilon_o} = 8{,}99 \cdot 10^9\, N \cdot m^2/C^2$$

$$G = 6{,}67 \cdot 10^{-11}\, N \cdot m^2/kg^2$$

Gravitational interaction is much weaker than the electrostatic interaction

PRINCIPLE OF SUPERPOSITION

$$\vec{\mathbf{F}}_{cał} = \sum_i \vec{\mathbf{F}}_i$$

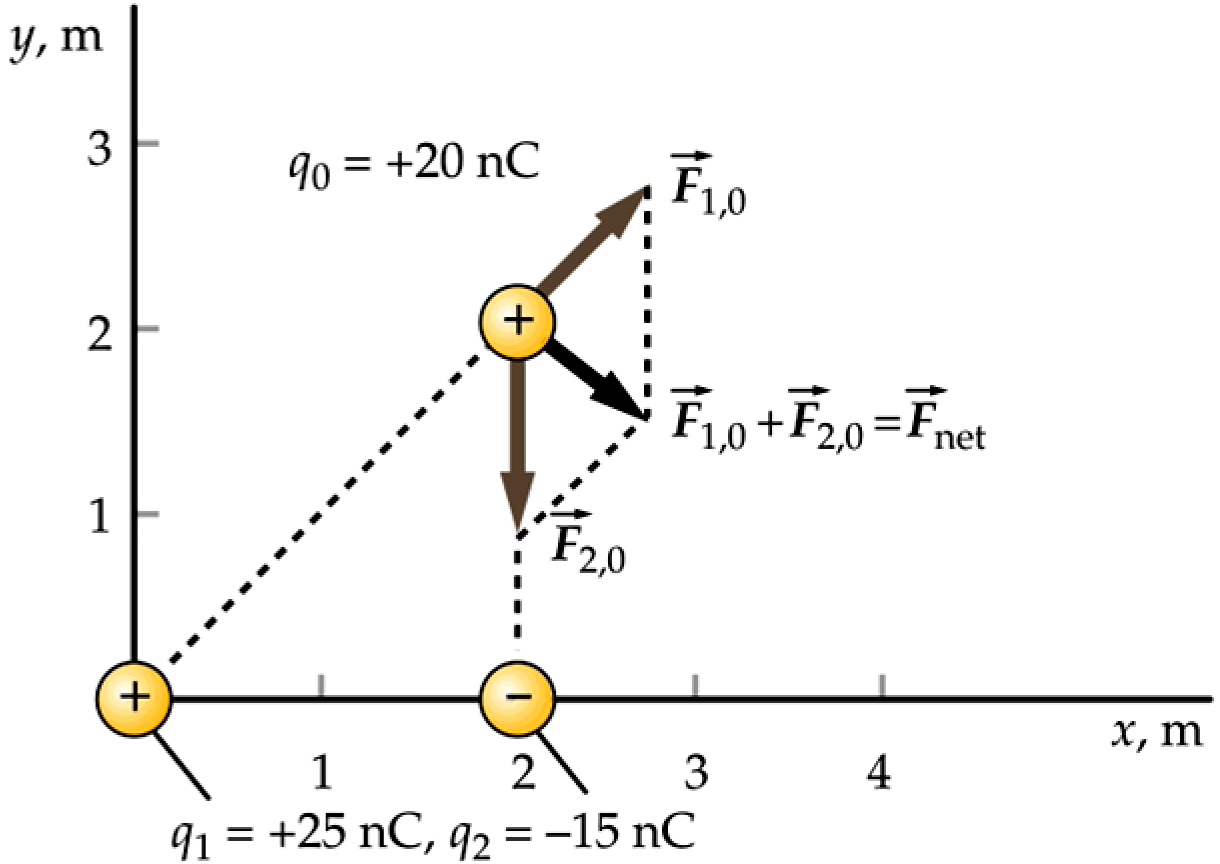


$$\vec{\mathbf{F}}_0 = \vec{\mathbf{F}}_R + \vec{\mathbf{F}}_L = \frac{kq_0(q_L - q_R)}{x^2}\hat{\mathbf{x}}$$

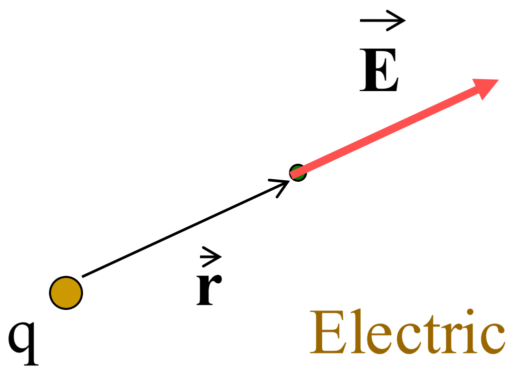
Charges q_L, q_0 and q_R have the same sign

Homework 9-1

Find the vector (its magnitude and direction) of the net force acting on the charge q_0



Definition of an electric field vector



$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q_0}$$

test charge $q_0 > 0$

Electric field vector due to a point (discrete) charge

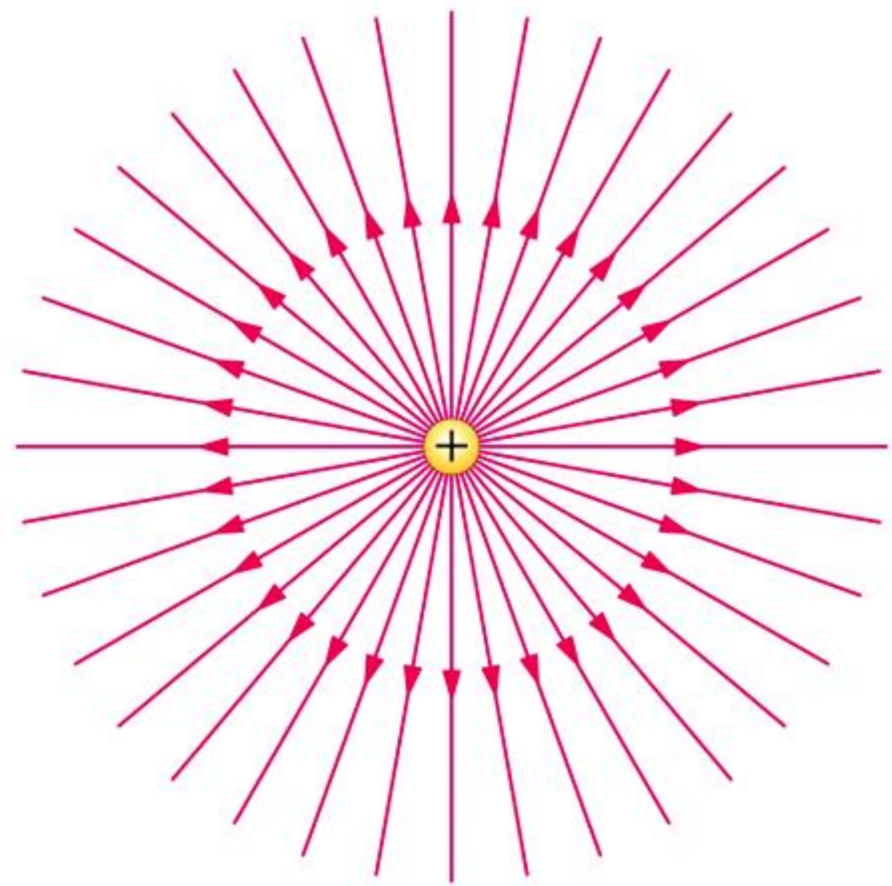
$$\vec{\mathbf{E}} = \frac{kq}{r^2}\hat{\mathbf{r}}$$

Electric field vector from multiple point charges (discrete distribution of charges)

$$\vec{\mathbf{E}} = \sum_i \vec{\mathbf{E}}_i = \sum_i \frac{kq}{r_i^{\,2}}\hat{\mathbf{r}}_i$$

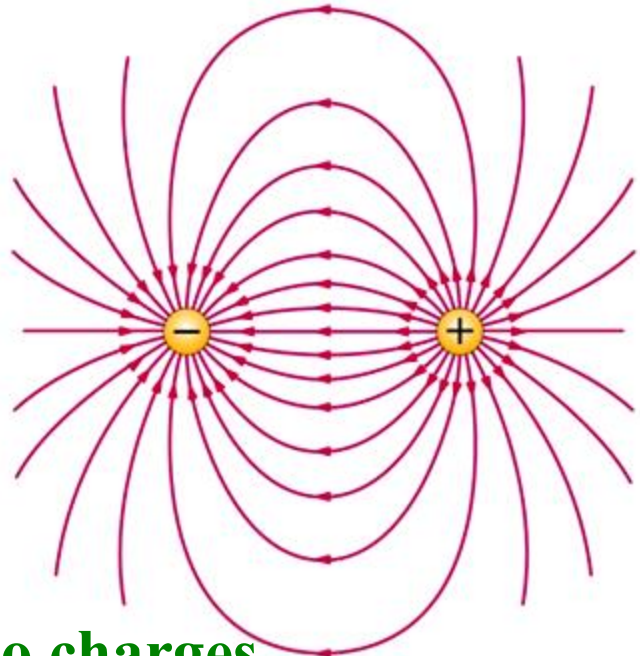
Field lines – **lines parallel to the electric field vector**

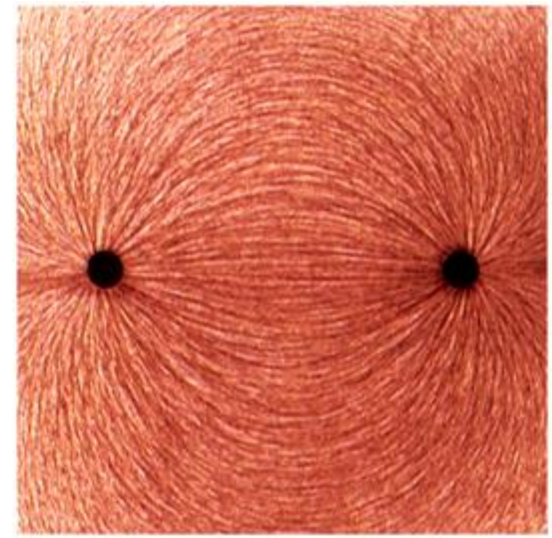
Electric field of a point charge (positive)



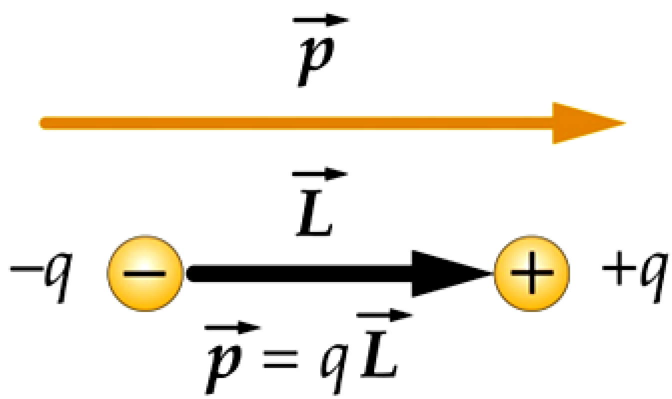
Spherical symmetry

Field lines





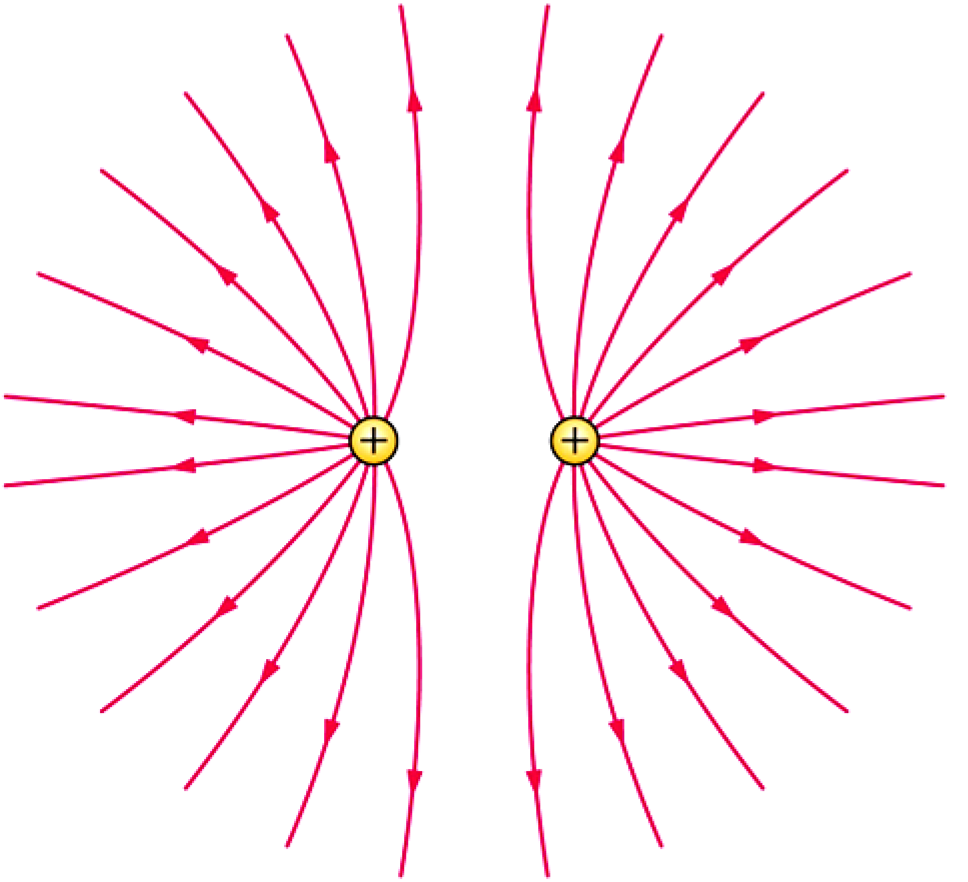
Electric dipole – two charges of the opposite sign placed at a small distance



Electric dipole moment

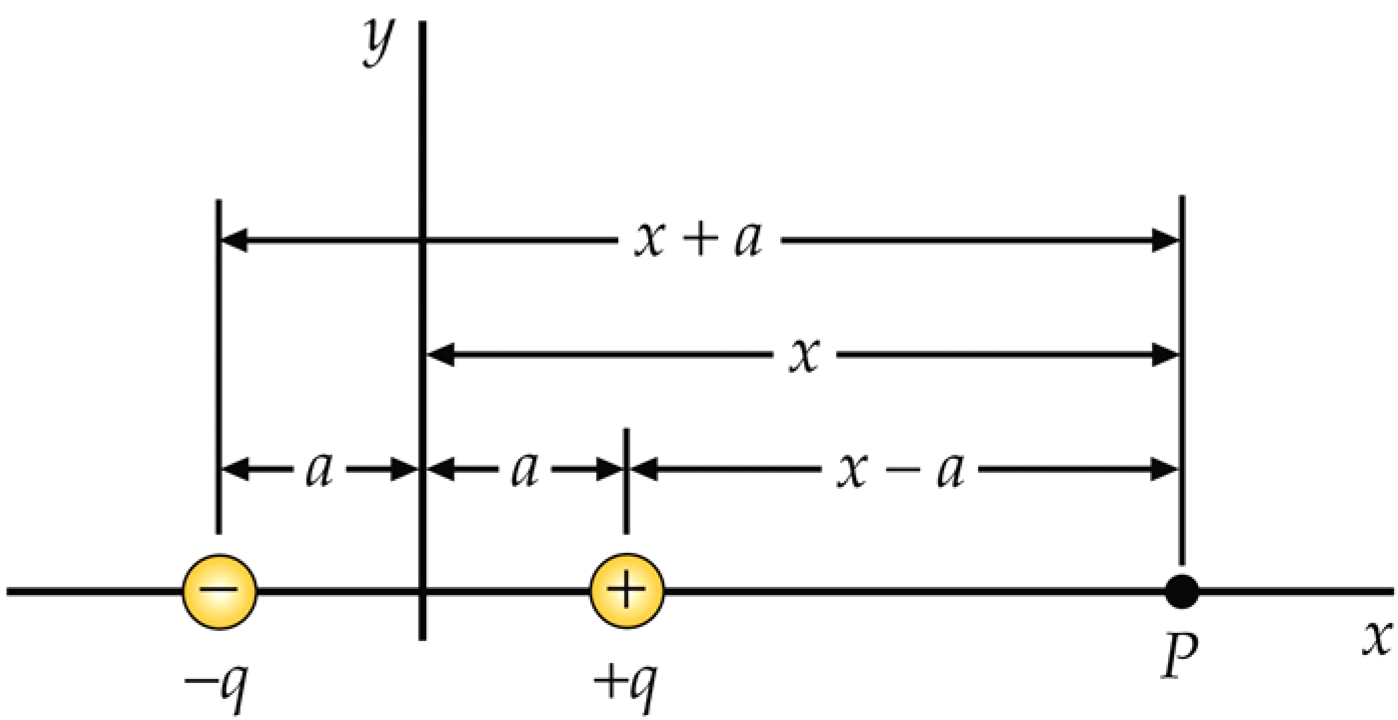
Field lines

Two point charges of the same sign



Homework 9-2

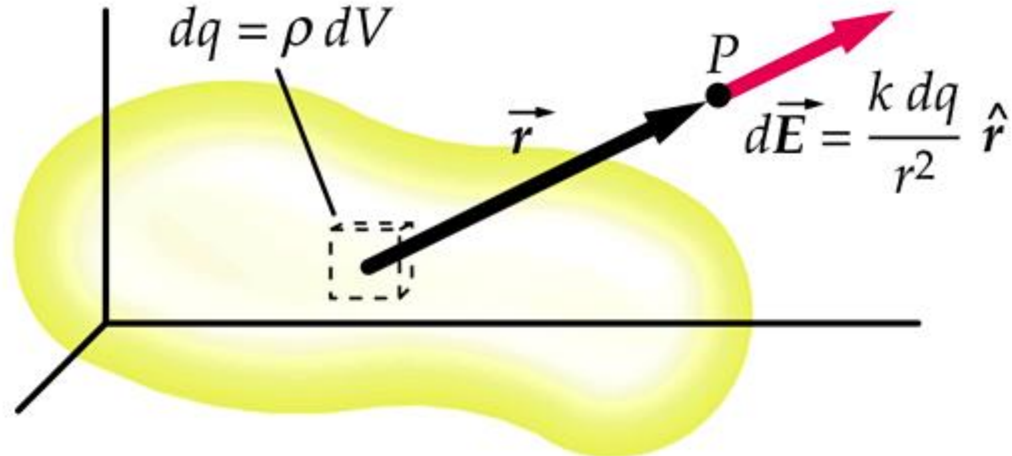
(a) Find the electric field vector at a point P for $x > a$
(b) Discuss the limiting case of $x >> a$ *(dipole approximation)*



Continuous charge distribution

For a charge, dq, the electric field vector at point P is given according to the Coulomb's law as for a point charge

$$d\vec{\mathbf{E}} = \frac{kdq}{r^2}\hat{\mathbf{r}}$$



In the case of discrete electric charges, the net electric field $\vec{\mathbf{E}}$ is a sum of the electric field vectors $\vec{\mathbf{E}}_i$ thus

$$\vec{\mathbf{E}} = \sum_{i} \vec{\mathbf{E}}_i$$

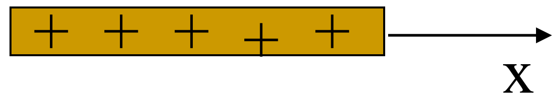
For a continuous charge distribution, the net field vector is given by an integral:

$$\vec{\mathbf{E}} = \int d\vec{\mathbf{E}} = \int_Q \frac{kdq}{r^2}\hat{\mathbf{r}}$$

Continuous charge distribution

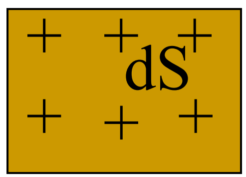
Depending of the dimensionallity:

•linear density of charge λ,



$$\lambda = \frac{dq}{dx}$$

•surface density of charge σ,



$$\sigma = \frac{dq}{dS}$$

•Volume density of charge ρ

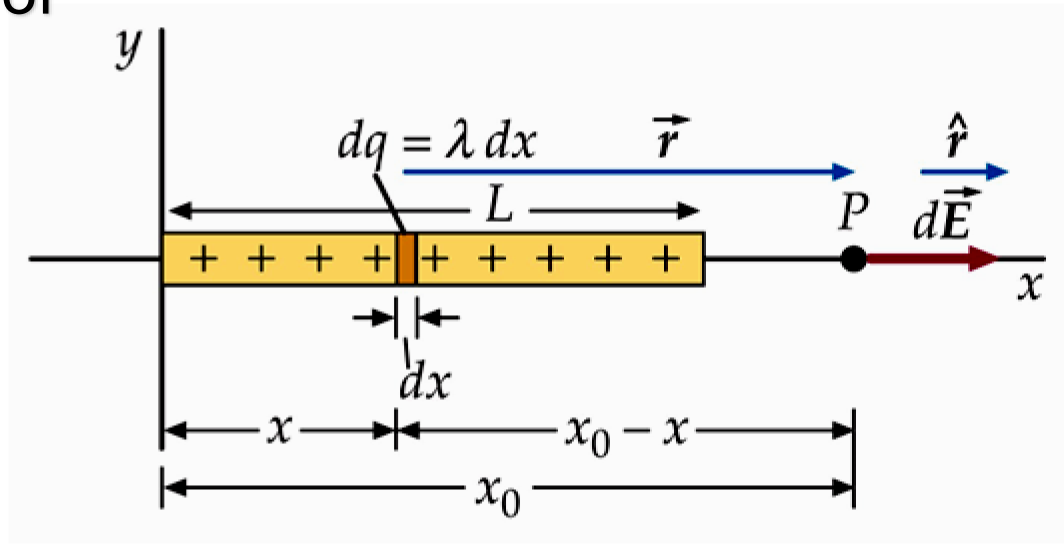
$$\rho = \frac{dq}{dV}$$

For a continuous charge distribution, depending on the type of charge density, the net force can be expressed as a linear, surface or volume integral

$$\vec{\mathbf{E}} = \int d\,\vec{\mathbf{E}} = \int_V \frac{k\,\rho\, dV}{r^2}\,\hat{\mathbf{r}}$$

Example 9-1 Linear charge distribution

Find the electric field vector at point P along the axis of linear charge distribution



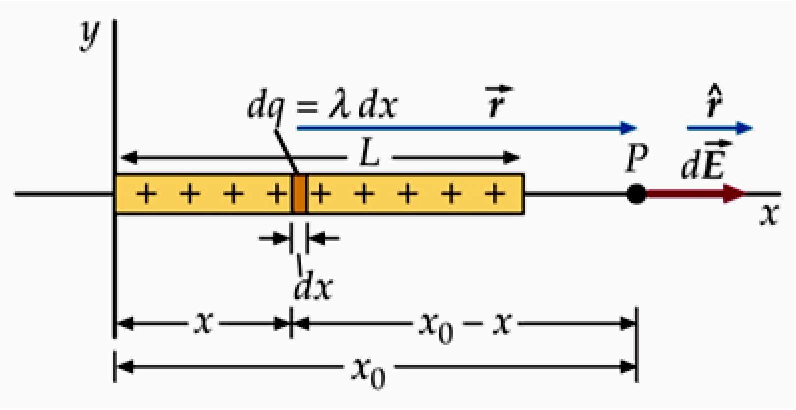
Coulomb's law

$$d\vec{\mathbf{E}}_x = \frac{kdq}{(x - x_o)^2}\hat{\mathbf{x}}$$

From the definition of linear charge density

$$dq = \lambda dx$$

$$d\vec{\mathbf{E}}_x = \frac{k\lambda d\,x}{(x - x_o)^2}\hat{\mathbf{x}}$$

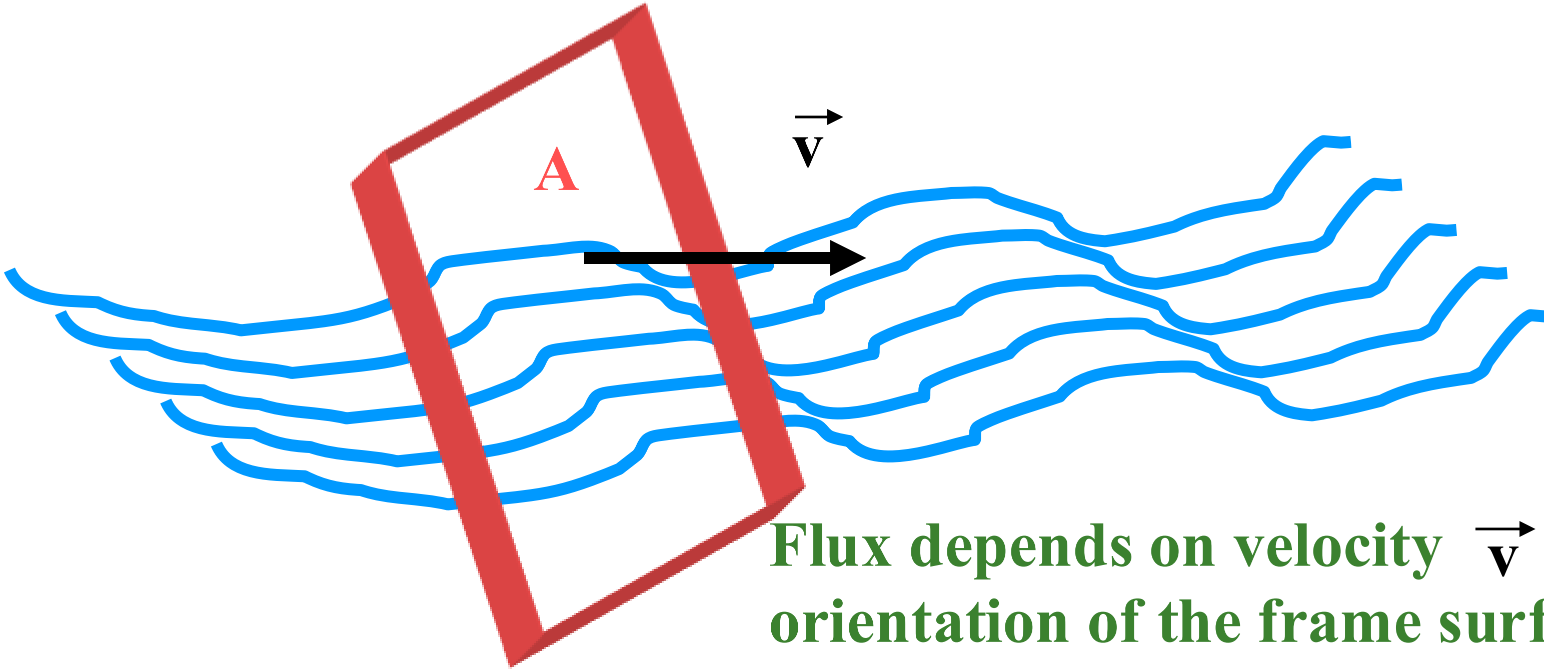


Net vector field is a sum of all contributions coming from the elemental charges dq:

$$E = \int dE_x = \int_0^L \frac{k\lambda dx}{(x_o - x)^2} \quad \boxed{= \frac{k\lambda L}{x_o(x_o - L)}}$$

FLUX

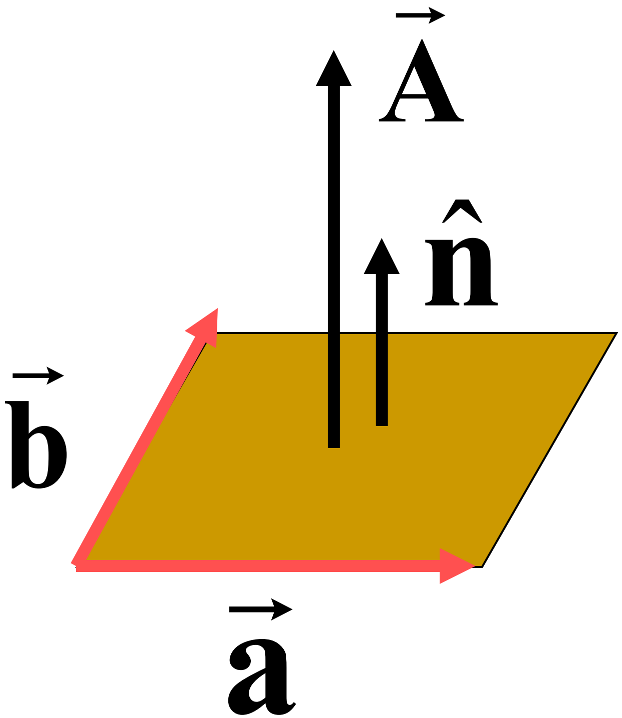
Φ - Flow rate (fluid=gas or liquid) through a surface A, i.e., the volume of fluid passing through a frame in a unit of time



Flux depends on velocity $\vec{v}$ and orientation of the frame surface

SURFACE AS A VECTOR $\vec{\mathbf{A}} = \mathbf{A}\hat{\mathbf{n}}$

$\hat{\mathbf{n}}$ - Unit vector perpendicular to the surface (normal)



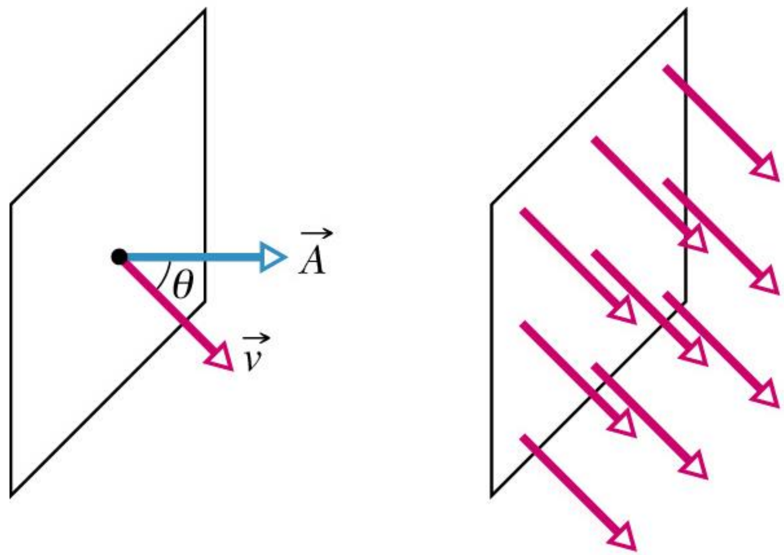
For a magnitude of the surface vector one can use:

$$\mathbf{A} = \left|\vec{\mathbf{a}} \times \vec{\mathbf{b}}\right|$$

Area of a parallelogram

FLUX OF VECTORIAL QUANTITY

velocity vector



- Flux of $\vec{\mathbf{v}}$ through a surface A

$$\Phi_{\vec{\mathbf{v}}} = \vec{\mathbf{v}} \circ \vec{\mathbf{A}} = \mathrm{vA}\cos\theta$$

- Flux of the electric field

$$\Phi_{\mathrm{E}} = \vec{\mathbf{E}} \circ \vec{\mathbf{A}}$$

Unit $1\mathrm{Nm}^2/\mathrm{C}$

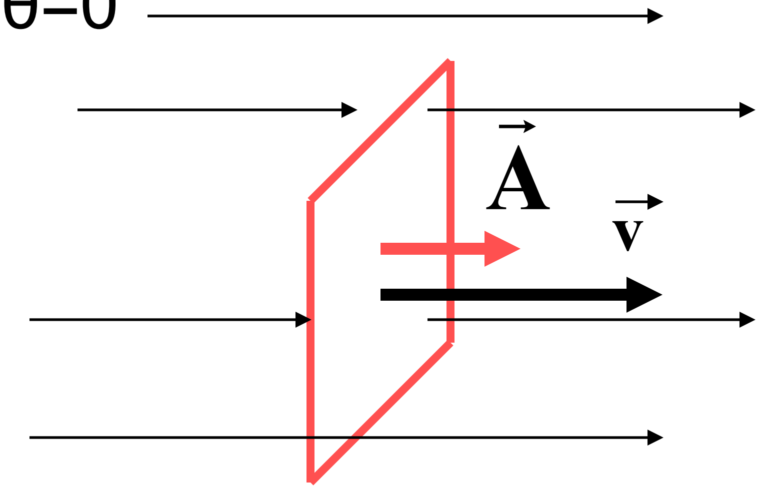
- Flux of magnetic field

$$\Phi_{\mathrm{B}} = \vec{\mathbf{B}} \circ \vec{\mathbf{A}}$$

Unit 1 Wb (weber) = 1 T (tesla) m^2

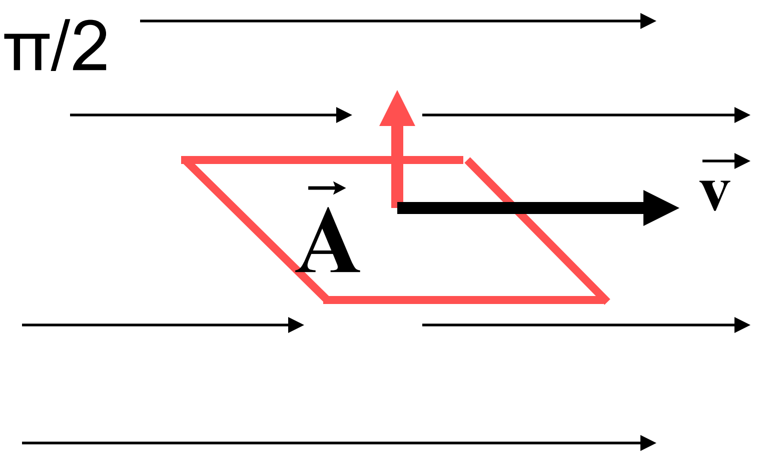
Particular cases:

- when θ=0



$$\Phi_{\vec{v}} max = vA$$

- when θ=π/2



$$\vec{A} \perp \vec{v}$$

$$\Phi_{\vec{v}} = 0$$

FLUX OF THE ELECTRIC FIELD – definition

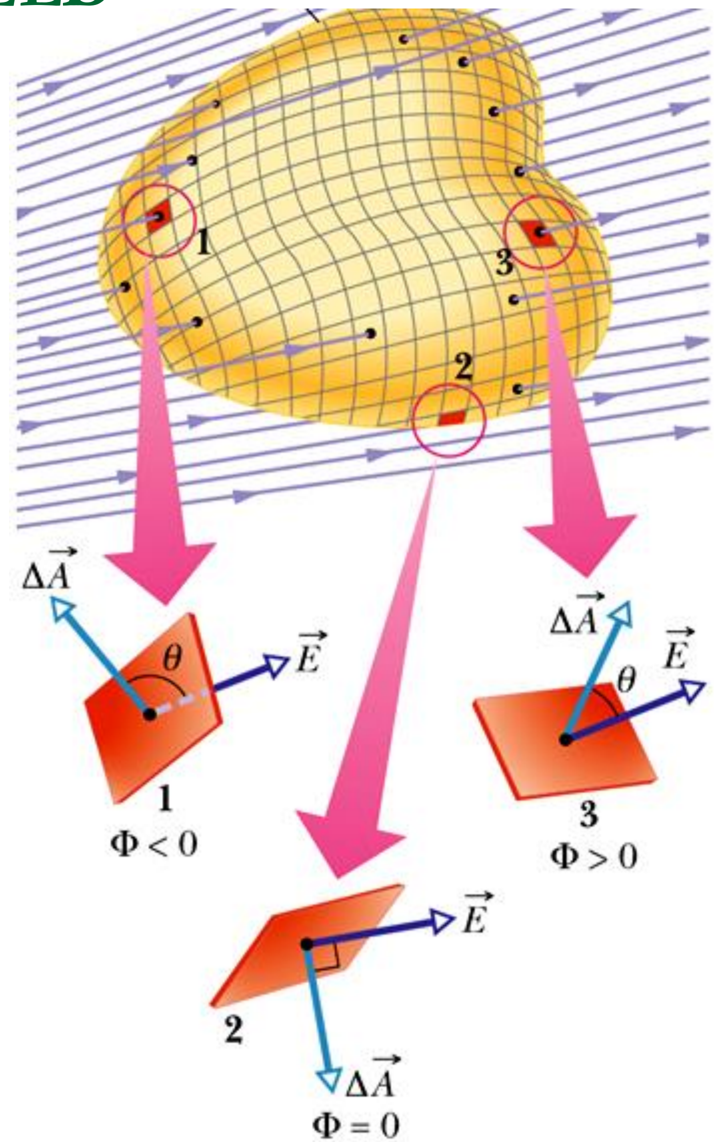
for any surface

$$\Delta\Phi_{\vec{\mathbf{E}}} = \vec{\mathbf{E}} \circ \Delta\vec{\mathbf{A}}$$

$$\Phi_{\vec{\mathbf{E}}} = \sum \vec{\mathbf{E}} \circ \Delta\vec{\mathbf{A}}$$

$$\boxed{\Phi_{\vec{\mathbf{E}}} = \int \vec{\mathbf{E}} \circ \mathrm{d}\vec{\mathbf{A}}}$$

definition

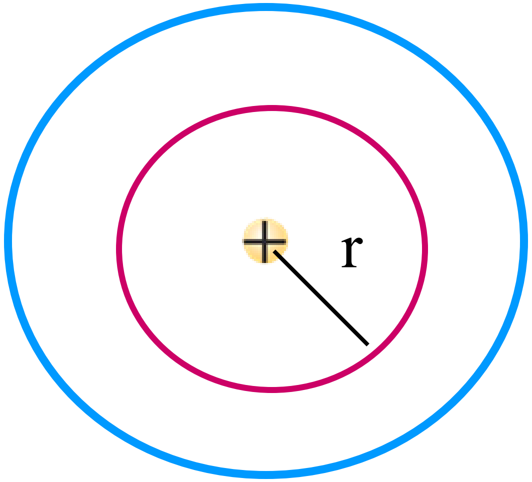


In Gauss' law there is a flux through a closed surface

$$\oint \vec{\mathbf{E}} \circ \mathrm{d}\vec{\mathbf{A}}$$

GAUSS' LAW

For a discrete (point) charge, $E \sim 1/r^2$



Estimated flux of the electric field through a spherical shell is Φ_E

$$\Phi_E = E\,A = \frac{1}{4\pi\varepsilon_o}\frac{Q}{r^2}4\pi r^2 = \frac{Q}{\varepsilon_o}$$

Surface of the sphere $A \sim r^2$

With the increasing distance from a source of the field, the surface A increases but the electric field E decreases in such a way that the flux of the field (the product of E and A) remains constant

GAUSS' LAW

<u>Total</u> flux of the electric field through a <u>closed surface</u> depends solely on the electric charge enclosed <u>within</u> this surface. This is a Gaussian surface.

$$\boxed{\Phi_E = \frac{Q_{in}}{\varepsilon_o}}$$

Gauss' law for the electric field in the integral form is one of the **<u>Maxwell equations</u>**

$$\boxed{\oint_S \vec{\mathbf{E}} \circ d\vec{\mathbf{A}} = \frac{Q_{wew}}{\varepsilon_o}}$$

Properties of Gaussian surface:

•Gaussian surface is hypothetical entity, mathematical, imaginary construction,

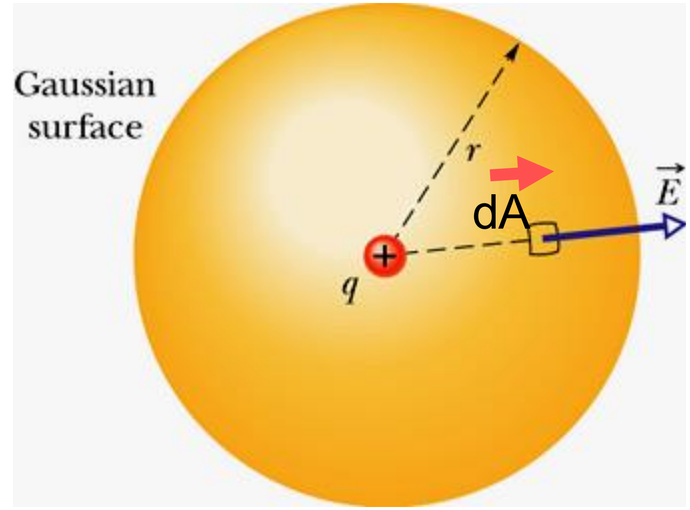
•It is an arbitrary closed surface, but in practice should take a shape related to the field symmetry,

•Gaussian surface should be drawn through a point at which we intend to calculate the magnitude of the electric field vector.

- Gauss' law can be used in calculations of the magnitude of the electric field vector when the distribution of charge is known or vice versa, i.e., knowing a field one can find the distribution of charge.
- Gauss' law is **always** valid, but should be used to calculate the magnitude of the electric field only when it has a well-defined symmetry (spherical, cylindrical).
- If you want to use effectively Gauss' law you need to **know something** about the electric field vector at the chosen Gaussian surface
- Using Gauss' law you can prove the equivalence of Gauss' and Coulomb's laws

From Gauss' to Coulomb's law

$$d\vec{A} = \hat{n}dA$$



1. Point charge q is surrounded by a Gaussian surface – a sphere of radius r (why?).

Because we know that an electric field due to a point charge has a spherical symmetry (What exactly does it mean?)

$\vec{E} \parallel \hat{n}$

cos 0 = 1

E = const. on the sphere surface

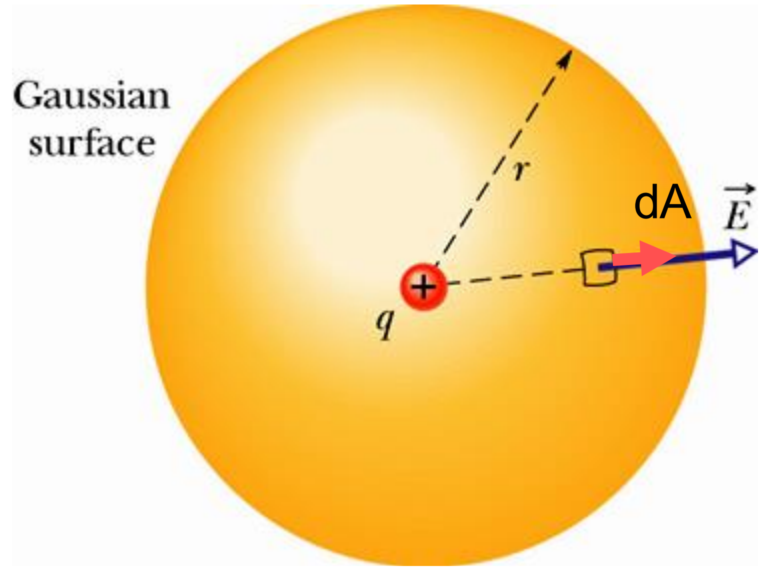
2. Calculate the total flux through the Gaussian surface

$$\Phi_E = \oint \vec{\mathbf{E}} \circ d\vec{\mathbf{A}} \quad = \oint E\cos\theta\, dA = \oint EdA$$

From Gauss' to Coulomb's law

$$d\vec{A} = \hat{n}dA$$

3. We use the fact, that E has a constant magnitude on the Gaussian surface



$$\Phi_E = E\oint dA = E(4\pi r^2)$$

4. From Gauss' law

$$\Phi_E = \frac{q}{\varepsilon_o}$$

5. Comparing two sides of equation:

$$E(4\pi r^2) = \frac{q}{\varepsilon_o} \Rightarrow E(r) = \frac{q}{4\pi\varepsilon_o r^2}$$

Coulomb's law

APPLICATIONS OF GAUSS' LAW

In practice the application of Gauss' law is limited to the particular cases – to certain symmetries:

a) field (uniform) due to an infinite in size, electrically charged surface (surface distribution of charge)

b) field (cylindrical symmetry) due to an infinitely long rod (linear distribution of charge) or a cylinder (surface distribution of charge – conducting cylinder, volume distribution of charge - non-conducting cylinder)

c) field (spherical symmetry) due to the charged solid sphere or a spherical surface

How to use Gauss' law?

1. Chose an appropriate Gaussian surface – fitting to the symmetry of the distribution of charge. Justify this choice. Make a drawing.
2. Calculate a flux of the electric field through the chosen Gaussian surface (*left hand side of Gauss' law*).
3. Find the charge enclosed <u>within</u> the Gaussian surface (*right hand side of Gauss' law*).
4. Compare two sides of Gauss' law and calculate the magnitude E of electric field vector.

Example 9-2. Electric field due to the infinite nonconducting sheet – planar symmetry

- Total flux of the electric field

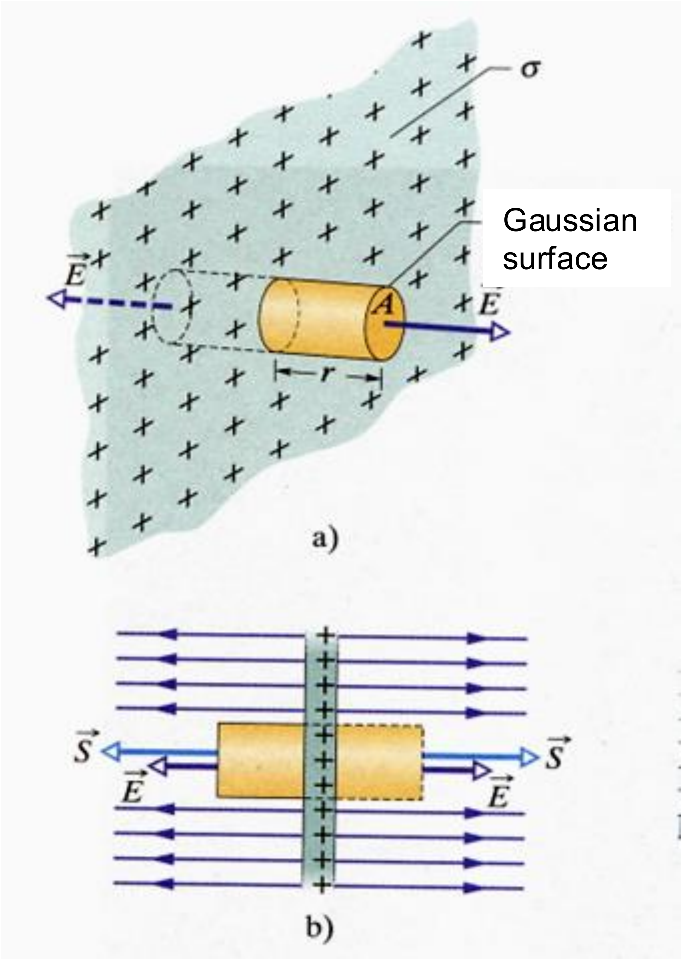
$$\Phi_E = 2ES$$

- From Gauss' law

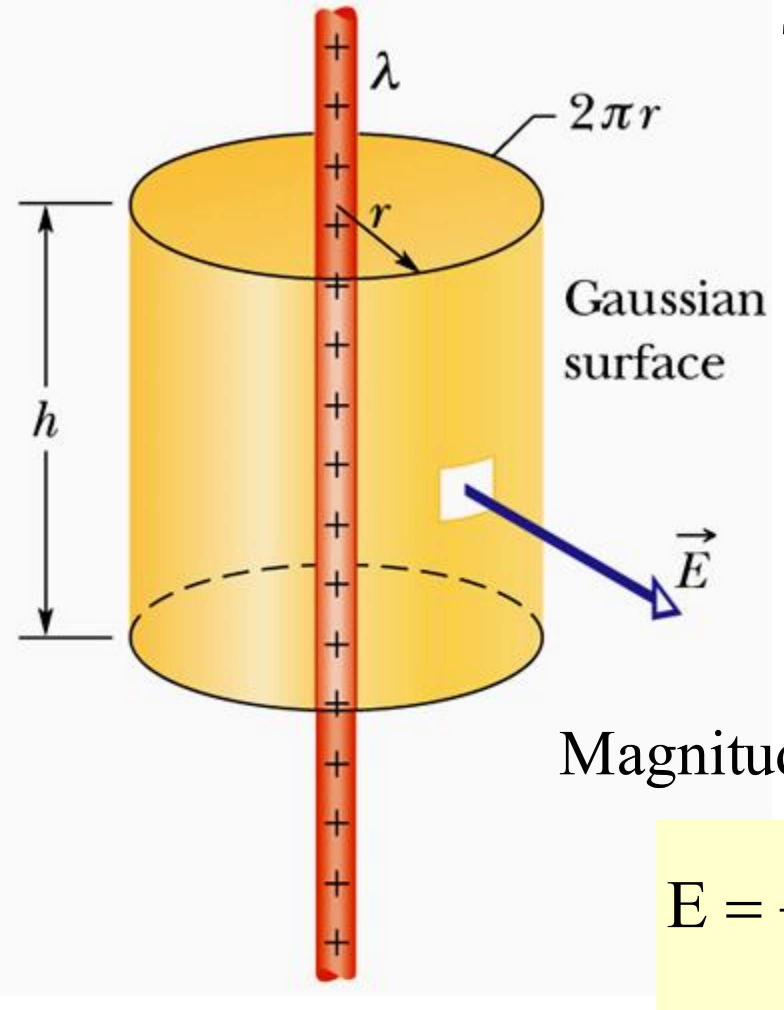
$$\Phi_E = \frac{Q}{\varepsilon_o} = \frac{\sigma S}{\varepsilon_o}$$

- Magnitude of the electric field vector

$$\boxed{E = \frac{\sigma}{2\varepsilon_o}}$$



Example 9-3. Electric field due to the linear charge density – infinite thin rod



Total flux of the electric field

$$\Phi_E = 2\pi rhE$$

Total charge enlosed by the Gaussian surface

$$\Phi_E = \frac{Q}{\varepsilon_o} = \frac{\lambda h}{\varepsilon_o}$$

Magnitude of the electric field vector

$$E = \frac{\lambda}{2\pi\varepsilon_o}\frac{1}{r}$$

cylidrical symmetry

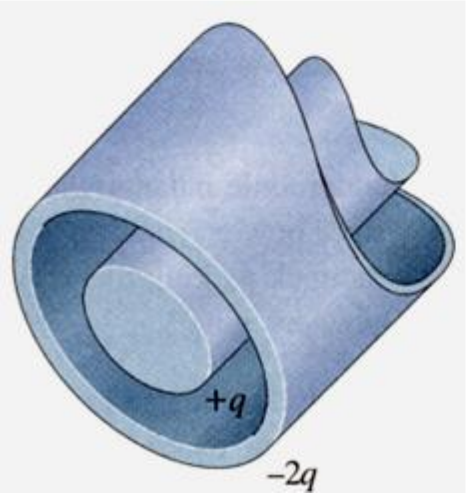
Homework 9-4.

Very long conducting rod (L - its length), charged with +q, is surrounded by a thin-walled coaxial conducting cylindrical shell of the same length L and a net charge -2q (as shown). Use the Gauss' law to determine:

(a) magnitude of the electric field vector outside the shell,

(b) distribution of charge in the shell,

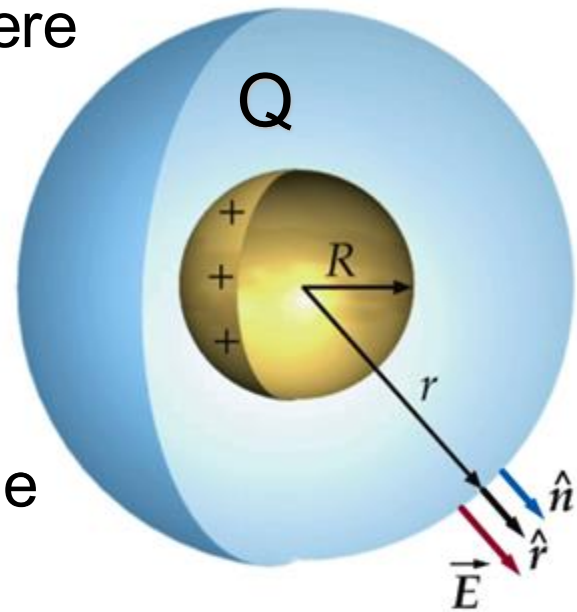
(c) magnitude of the electric field vector in the region between the rod and the shell



Example 9-4. Electric field in the case of a spherical symmetry of charge – spherical shell

Total flux through a Gaussian surface (a sphere of radius r) is:

$$\Phi_E = 4\pi r^2 E$$



From Gauss' law: $\Phi_E = \dfrac{Q}{\varepsilon_o}$

Net charge Q is located at the surface of the sphere of radius R

From the point of view of charge distribution we will consider two cases:

- r>R

$$E = \frac{Q}{4\pi\varepsilon_o r^2}$$

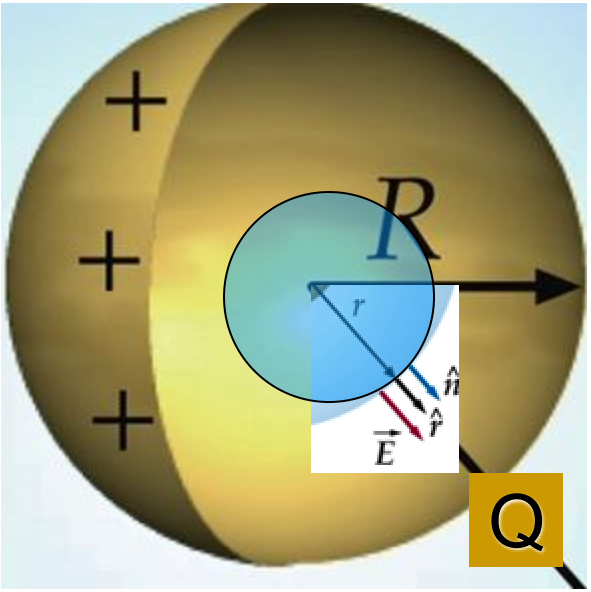
The electric field outside the empty spherical shell is the same as if the net charge were concentrated at its center.

Example 9-4 continued

- r<R

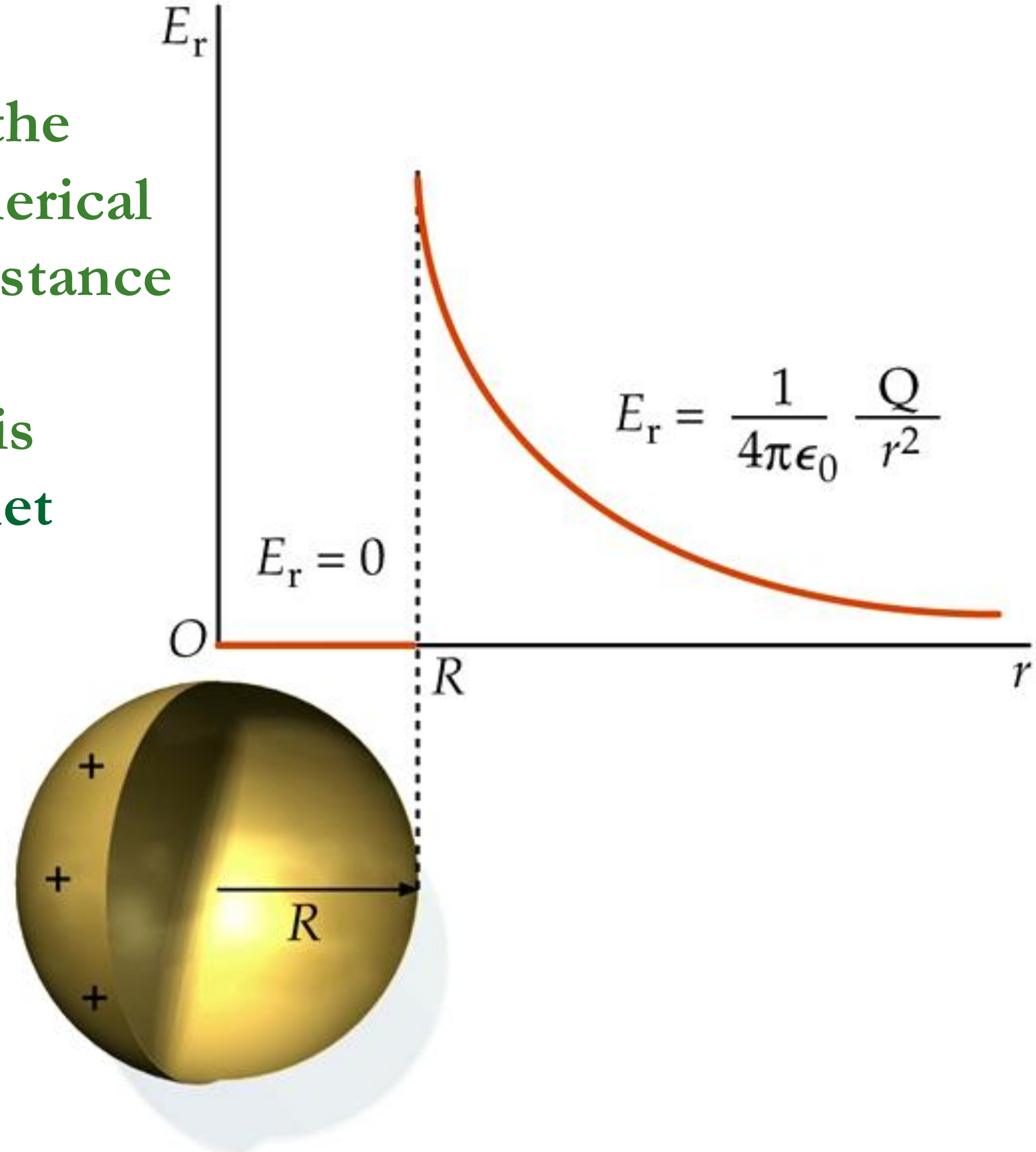
  inside Gaussian surface, i.e., a sphere of radius r, there is no charge

  $Q=0$, $\Phi_E=0$ and thus $E=0$



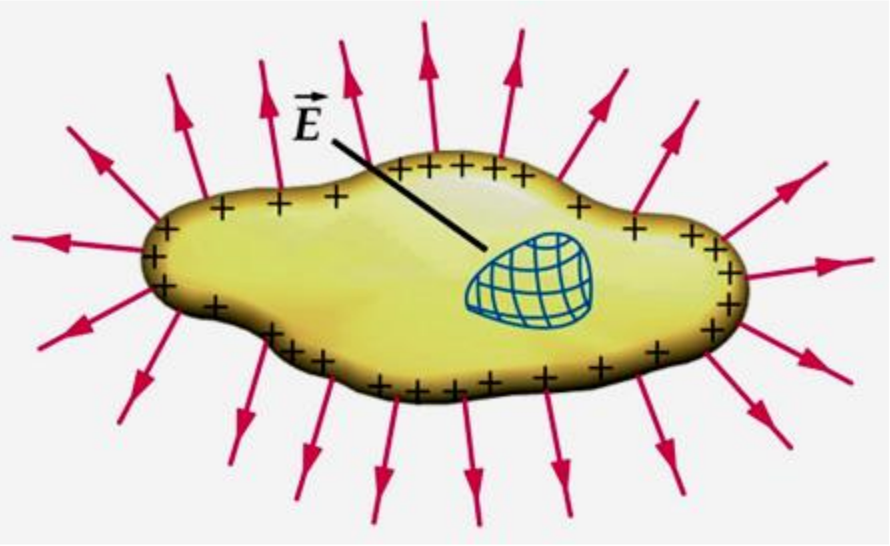
Electric field inside a charged spherical shell is zero.

Radial component E_r of the electric field due to a spherical shell as a function of a distance r from its center,
R – shell diameter, shell is uniformly charged (Q - net charge)



Electric field due to a conductor

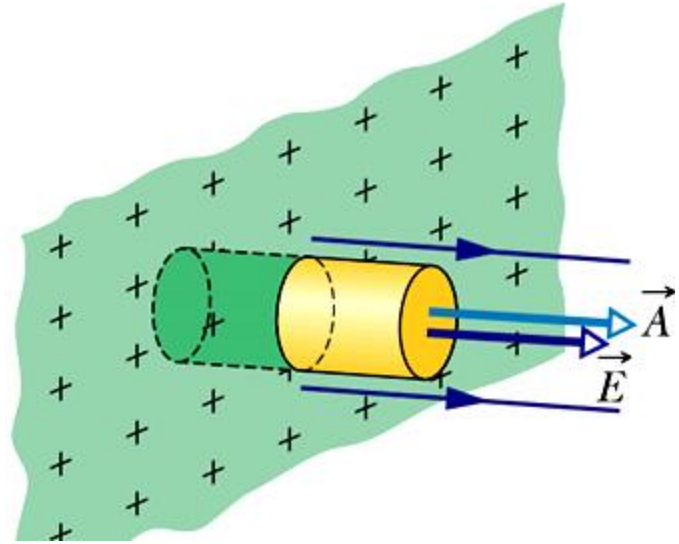
- Charge is distributed over the surface, only
- Inside the conductor Q=0, hence E=0
- At the surface of the conductor, the vector of the electric field $\vec{\mathbf{E}}$ is perpendicular to it



Electric field at the conductor surface

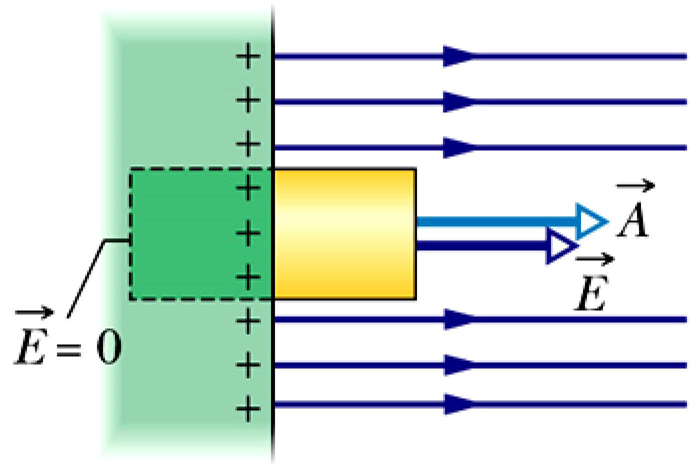
- Total flux through the Gaussian surface

$$\Phi_E = EA$$



- From Gauss' law

$$\Phi_E = \frac{Q}{\varepsilon_o} = \frac{\sigma A}{\varepsilon_o}$$



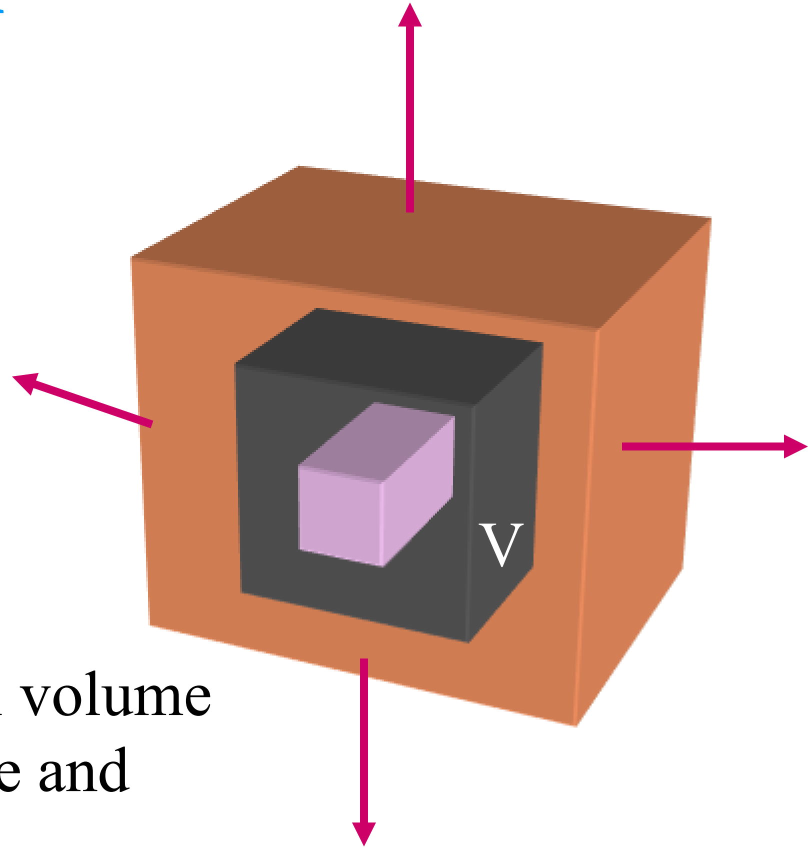
- Magnitude of the electric field vector

$$E = \frac{\sigma}{\varepsilon_o}$$

Flux and divergence operator

Definition of divergence

$$\operatorname{div}\,\vec{\mathbf{E}} = \lim_{V\to 0}\frac{\oint \vec{\mathbf{E}}\circ d\,\vec{\mathbf{A}}}{V}$$



$\operatorname{div}\,\vec{\mathbf{E}}$ is, in the limit of infinitely small volume V, flux emerging from the source and determines its efficiency

Gauss-Ostrogradsky theorem

$$\oint_S \vec{\mathbf{E}}\circ d\,\vec{\mathbf{A}} = \oiiint_V \operatorname{div}\,\vec{\mathbf{E}}\, dV$$

GAUSS' LAW IN DIFFERENTIAL FORM

From Gauss-Ostrogradsky theorem:

$$\oint_S \vec{\mathbf{E}} \circ d\vec{\mathbf{A}} = \oiiint_V \operatorname{div} \vec{\mathbf{E}}\, dV$$

Gauss' law in the integral form:

$$\oint_S \vec{\mathbf{E}} \circ d\vec{\mathbf{A}} = \frac{Q_{wew}}{\varepsilon_o} = \frac{1}{\varepsilon_o} \oiiint_V \rho dV$$

Comparing functions under integrals:

$$\boxed{\operatorname{div} \vec{\mathbf{E}} = \frac{\rho}{\varepsilon_o}}$$

charge density

POTENTIAL

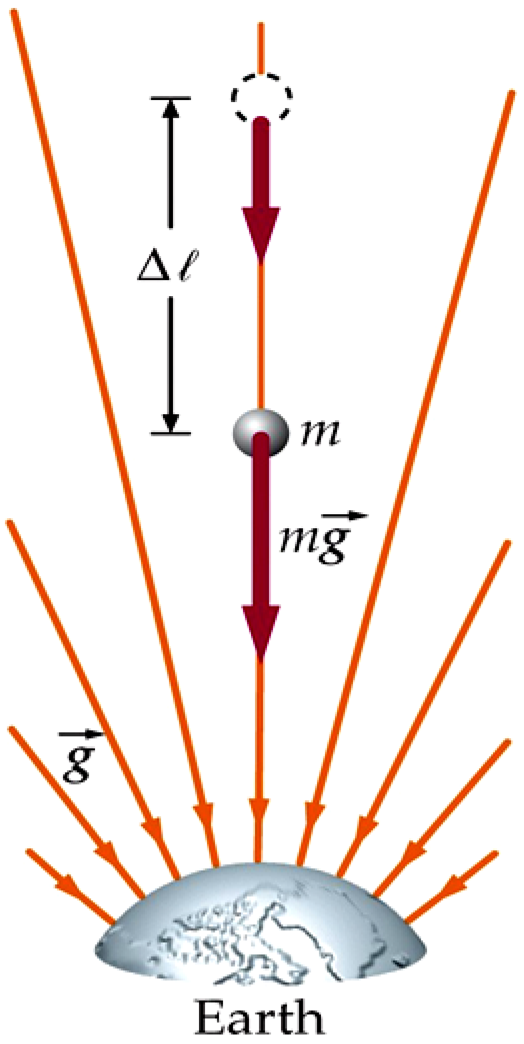
- Field vector – always exists

$$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q_o}$$ definition

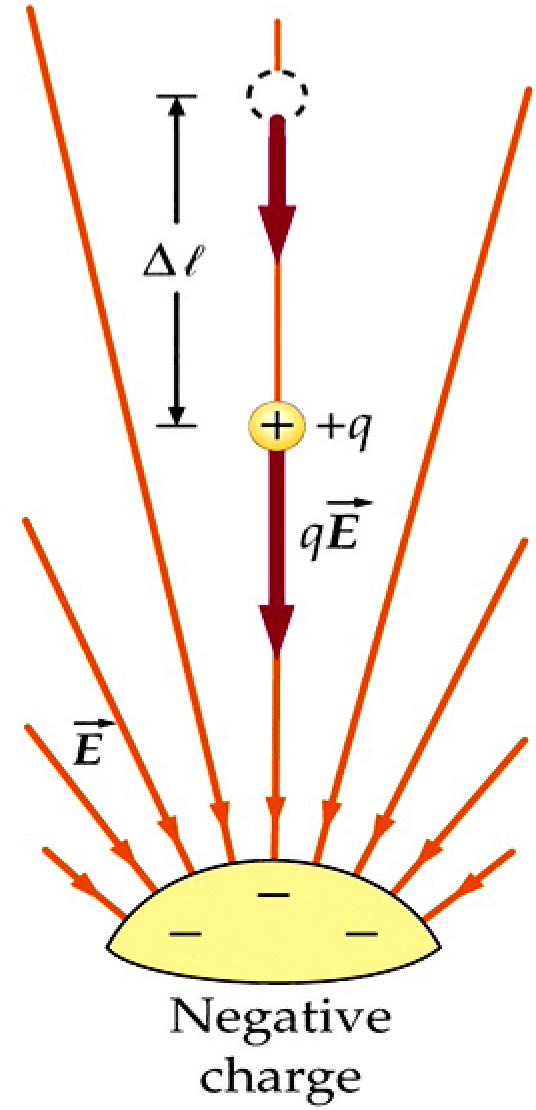
- Scalar potential – can be defined only in the case of conservative fields

$$V = \frac{E_p}{q_o}$$ definition

Quantities that characterize:	interaction between point charges	electrostatic field
force $\vec{\mathbf{F}}$ potential energy E_p	$\vec{\mathbf{F}} = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}$ $E_p = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r}$	
field vector $\vec{\mathbf{E}}$ potential V		$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}}{q_o}$ $V = \frac{E_p}{q_o}$



Gravitational field



Electrostatic field

Relationship between potential and electric field vector

For a conservative force: $$dE_p = \underbrace{-\vec{F} \circ d\vec{l}}_{\text{work } dW} = -q_o \vec{E} \circ d\vec{l}$$

From potential definition :

$$dV = \frac{dE_p}{q_o}$$

$$dV = -\vec{E} \circ d\vec{l}$$

$$\boxed{V_b - V_a = -\int_a^b \vec{E} \circ d\vec{l}}$$

$$\boxed{\vec{E} = -\,\text{grad } V}$$

$$V_b - V_a = -\int_a^b \vec{\mathbf{E}} \circ d\,\vec{\mathbf{l}}$$

$$\Delta V = V_b - V_a = -\frac{W}{q_o}$$

Potential difference ΔV between two points is equal to the negative work W done by the electrostatic force, when the unit charge is displaced from one point to another

Potential difference is called voltage U=ΔV

Units

Based on the following equation: $$V = \frac{E_p}{q_o}$$

- unit of potential is 1 V = 1 J/C
- 1 eV is a unit of energy at the atomic scale

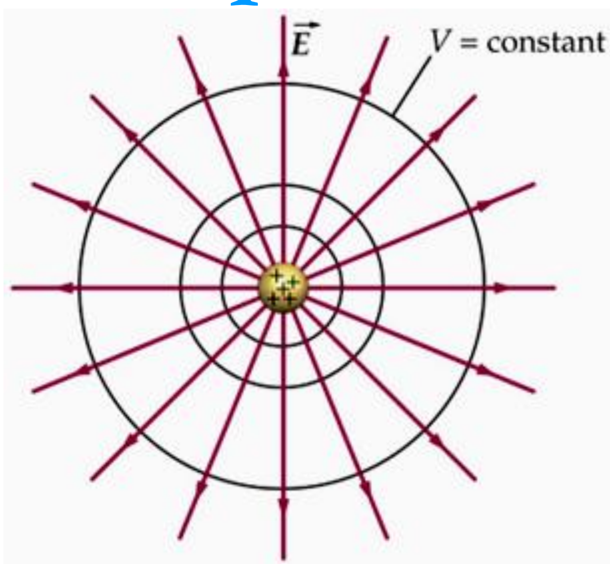
One electron-volt (eV) is the energy equal to the work required to move a single elementary charge *e*, such as that of the electron or the proton, through a potential difference of exactly one volt.

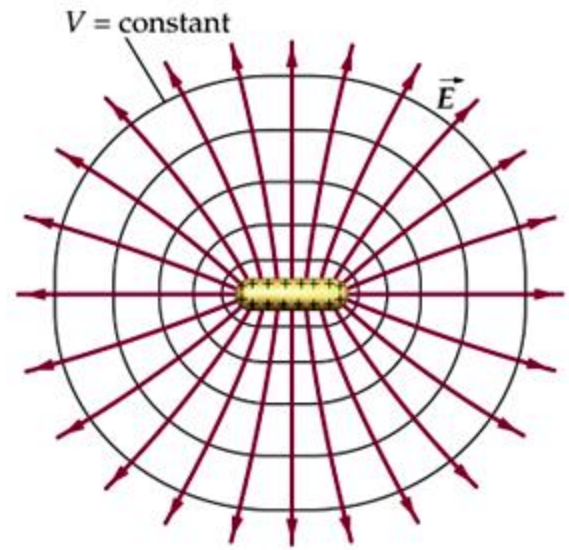
1 eV= e (1V) = (1.6 $\cdot 10^{-19}$ C) (1 J/C) = 1.6 $\cdot 10^{-19}$ J

Based on the equation $$V_b - V_a = -\int_a^b \vec{\mathbf{E}} \circ d\vec{\mathbf{l}}$$

• we can define a unit of the electric field 1 V/m

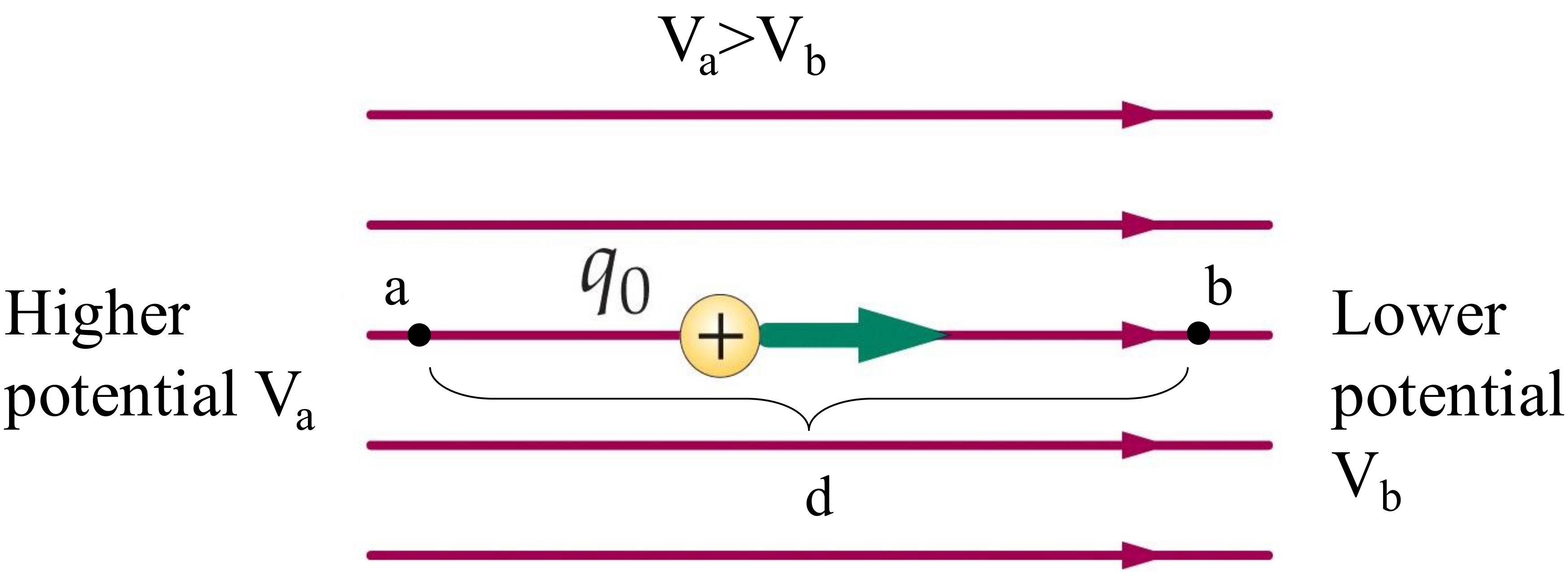
Equipotential surfaces - surfaces of constant potential





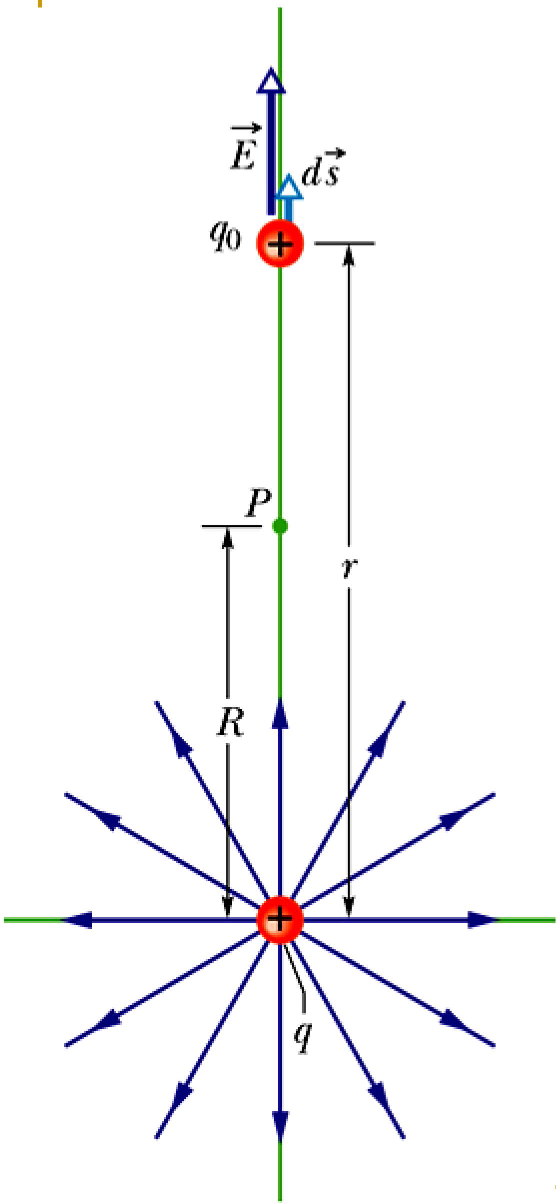
Adjacent points that have the same electric potential form an equipotential surface, which can be either an imaginary or a real, physical surface. No net work is done on a charged particle by the electric field when the particle moves between two points on the equipotential surface. Equipotential surfaces are always perpendicular to the electric field lines.

Potential of the uniform electric field



$$V_b - V_a = -\int_a^b \vec{\mathbf{E}} \circ d\,\vec{\mathbf{l}} = -\int_a^b E\,dl = -E\int_a^b dl = -E(b-a) = -Ed$$

Potential due to a point charge



We move a positive test charge q_o from P to infinity

$$\vec{\mathbf{E}} \circ d\,\vec{\mathbf{s}} = E\,ds\cos\theta$$

$$V_\infty - V_P = -\int_R^\infty \vec{\mathbf{E}} \circ d\,\vec{\mathbf{s}} = -\int_R^\infty E dr \qquad E = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}$$

We assume $V_\infty=0$

$$\boxed{V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}}$$

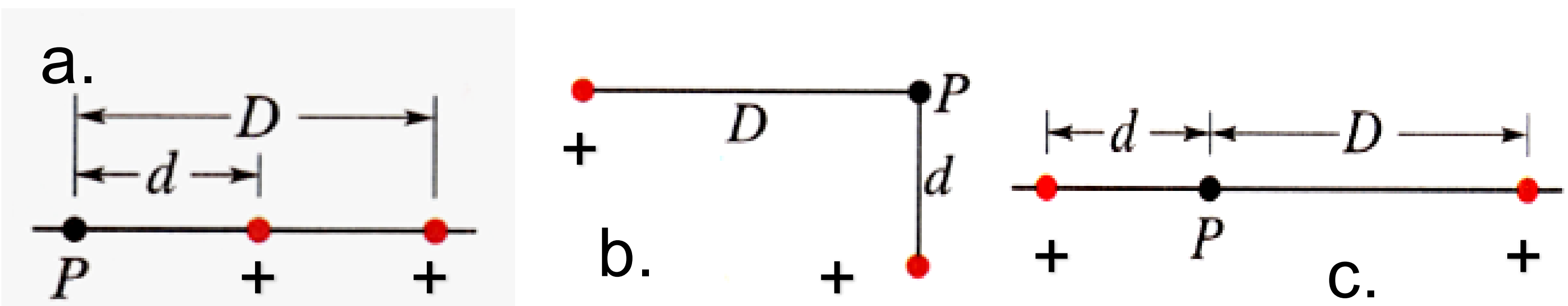
The sign of V is the same as the sign of q charge being a source of the field.

Potential due to a discrete distribution of charge

Net potential V of a system of n point charges q_i can be calculated applying <u>principle of superposition</u>

$$V = \sum_{i=1}^{n} V_i = \frac{1}{4\pi\varepsilon_o} \sum_{i=1}^{n} \frac{q_i}{r_i}$$

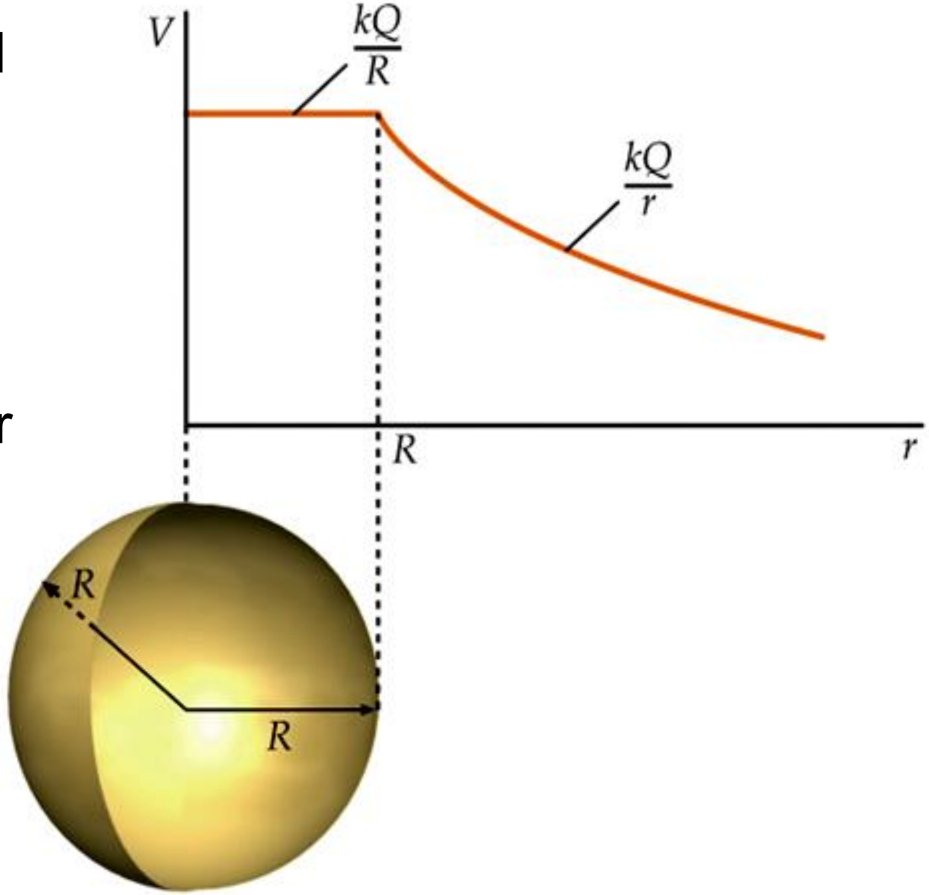
Homework 9-5 Drawing below illustrates three systems containing two protons. Rank these systems according to the decreasing net electric potential at the point P



Potential due to continuous charge distribution

Surface charge Q of the spherical shell of R radius produces at a distance r<R from its center, E=0, that is why the potential V is constant, independent of r.

For r>R, V decreases with r as 1/r



Homework 9-6 Demonstrate by calculations that the potential V(r) due to a spherical shell, behaves as shown above.

CAPACITANCE

- Definition $$C = \frac{Q}{\Delta V}$$

The unit of capacitance is 1F (one farad). In practice we use μF, pF, nF

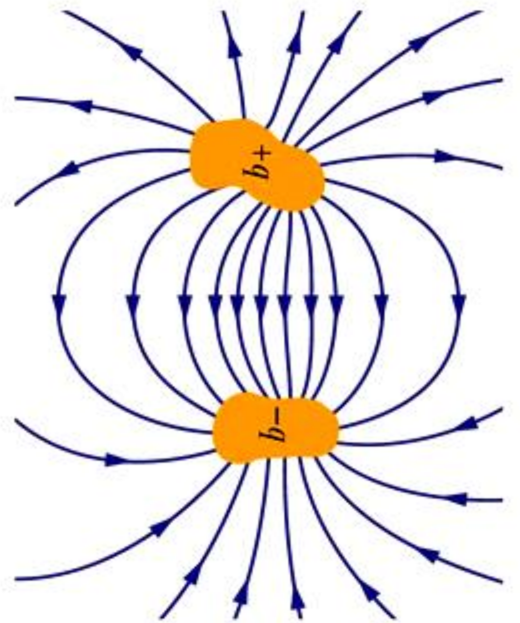
There is an analogy between the capacitor with charge q and the rigid box of volume (capacity) $\mathcal{V}$, containing n moles of ideal gas:

$$n = \frac{\mathcal{V}}{RT}p \qquad q = C\Delta V$$

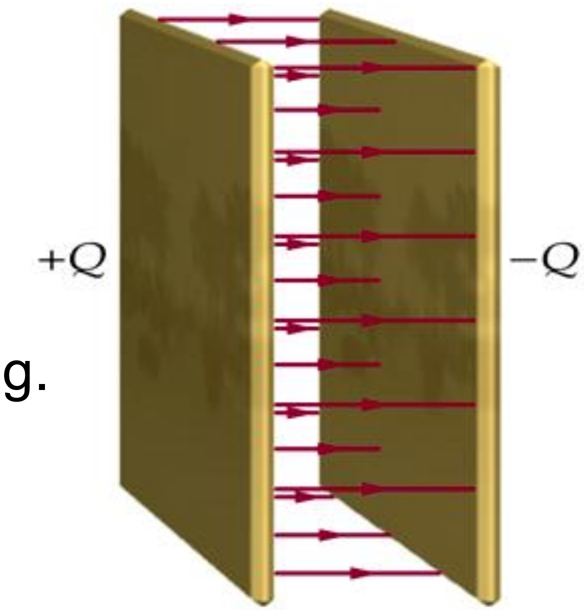
Capacitance C plays a similar role as a volume $\mathcal{V}$ at a constant temperature T. The capacitance is a measure of how much charge must be put on the conductor to produce a certain potential difference.

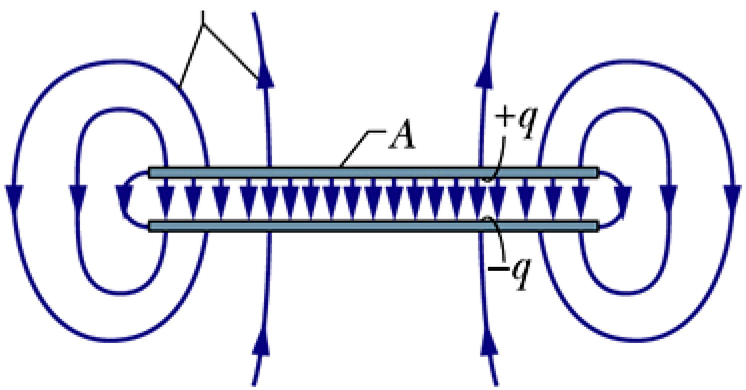
Capacitor

- Stores the electrical energy
- Serves as an important element of the electrical circuit



Examples of capacitors, e.g. a parallel-plate capacitor

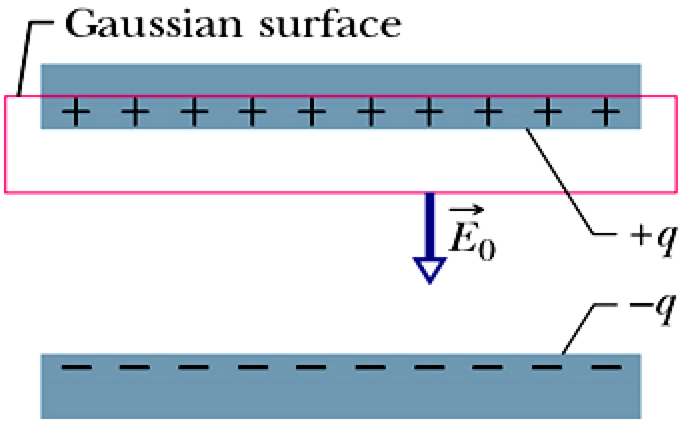




Parallel-plate capacitor

$$E_o = \frac{\sigma}{\varepsilon_o} = \frac{q}{\varepsilon_o A}$$

$$q = E_o \varepsilon_o A$$



For a uniform electric field we have:

$$V = E_o d$$

$$C = \frac{\varepsilon_o A}{d}$$

From a definition of capacitance

Capacitance depends on the geometry of the plates and not on their charge or potential difference, A-area of a plate, d-distance between the plates

Example 9-5: Estimate the area of the electrode of the parallel-plate capacitor, if you want to obtain the capacitance of C=1 F assuming a distance between plates of d=1 mm.

$$C = \frac{\varepsilon_o A}{d} \quad \Rightarrow \quad A = \frac{Cd}{\varepsilon_o}$$

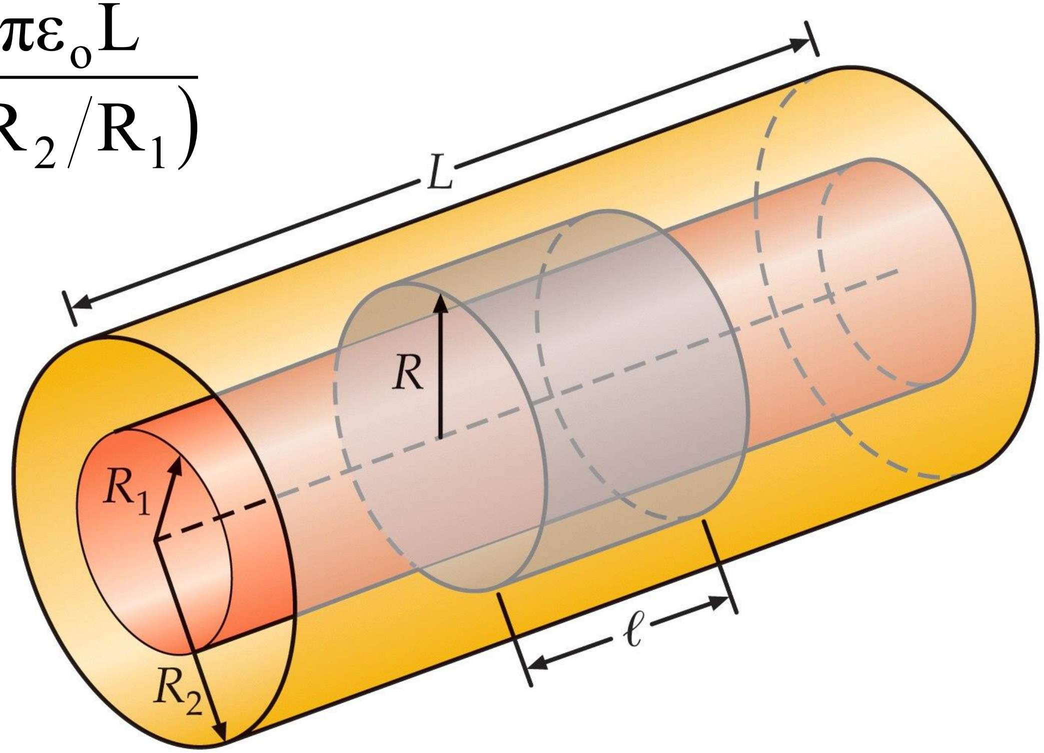
$$A = \frac{1\,\mathrm{F} \cdot 10^{-3}\,\mathrm{m}}{8{,}85 \cdot 10^{-12}\,\mathrm{C^2/N \cdot m^2}} = 1.13 \cdot 10^{8}\,\mathrm{m}^2$$

Draw your own conclusion!!!

Homework 9-7

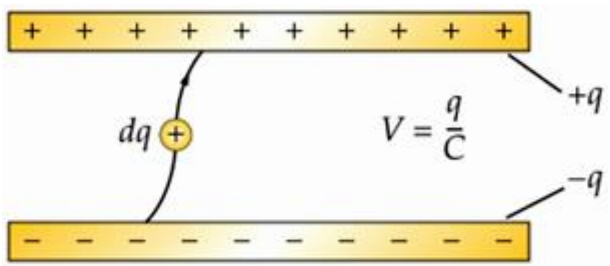
Prove that the capacitance of cylindrical capacitor is given by:

$$C = \frac{2\pi\varepsilon_o L}{\ln\left(R_2/R_1\right)}$$



Energy stored in an electric field

Work W must be done by an external agent to load a capacitor. This work is stored <u>as potential electric energy</u> E_E, in the field between two plates.

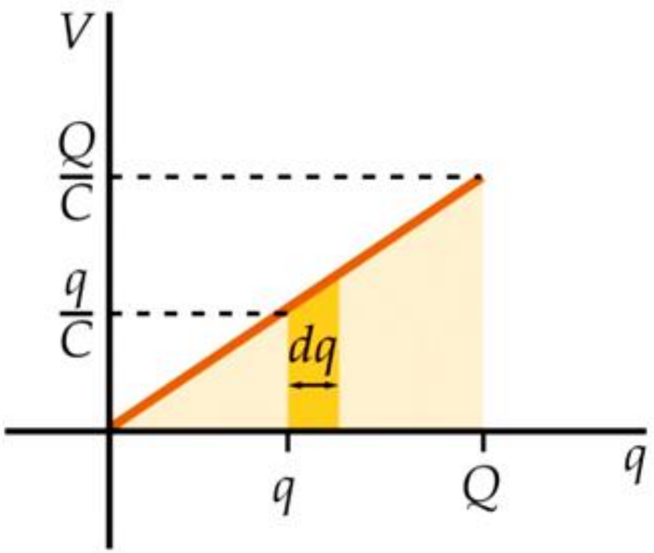


Elemental work dW done when the additional charge dq is transferred from one plate of a capacitor to another equals:

$$dW = Vdq = \frac{q}{C}dq$$

$$W = \int dW = \int \frac{q}{C}dq = \frac{1}{2}\frac{Q^2}{C}$$

$$W = E_E = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV = \frac{1}{2}CV^2$$



Energy density

Energy density u_E is a potential energy per unit volume and can be easily calculated for a parallel-plate capacitor

$$u_E = \frac{E_E}{Ad} = \frac{CV^2}{2Ad} \qquad C = \varepsilon_o \frac{A}{d}$$

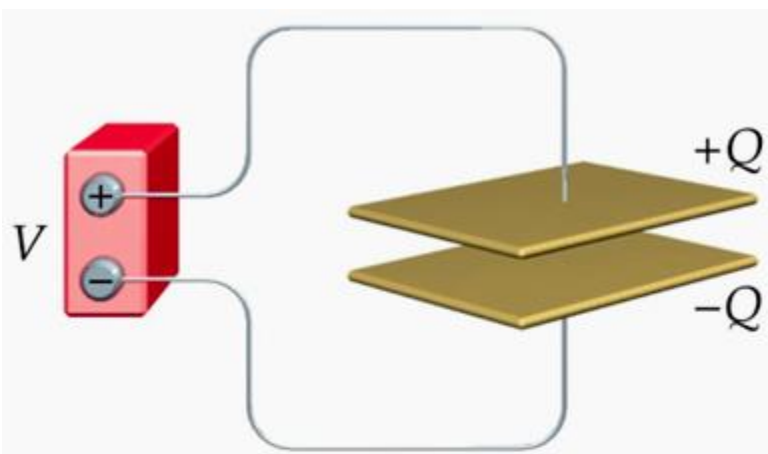
$$u_E = \frac{1}{2}\varepsilon_o \left(\frac{V}{d}\right)^2$$

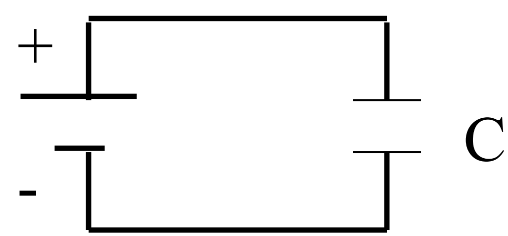
$$\boxed{u_E = \frac{1}{2}\varepsilon_o E^2}$$

This formula holds generally, whatever may be the source of the electric field

Capacitor in a circuit

Example 9-6: Capacitor of a capacitance C=6 μF is connected to 9 V battery as shown below. What charge is collected on its plates?





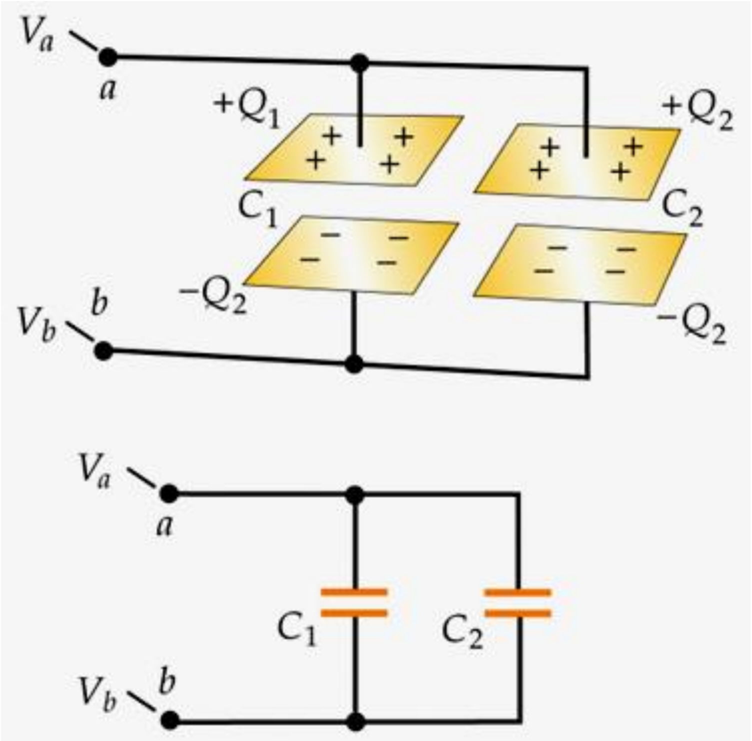
$Q = CV = 54\ \mu C$

Capacitors in parallel

$$V_a - V_b = U_1 = U_2 = U$$

$$\frac{Q_1}{C_1} = \frac{Q_2}{C_2} = \frac{Q}{C} = U$$

$$Q = Q_1 + Q_2 = C_1U + C_2U = (C_1 + C_2)U$$



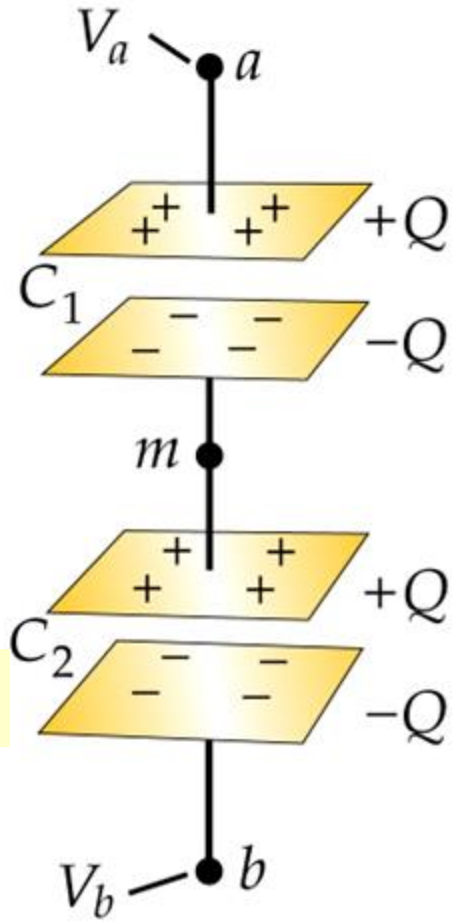
$$C = C_1 + C_2$$

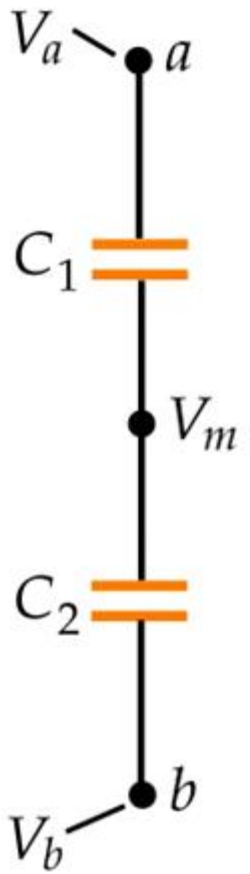
Equivalent capacitance C

Capacitors in series

$$U = U_1 + U_2 = \frac{Q}{C_1} + \frac{Q}{C_2}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$



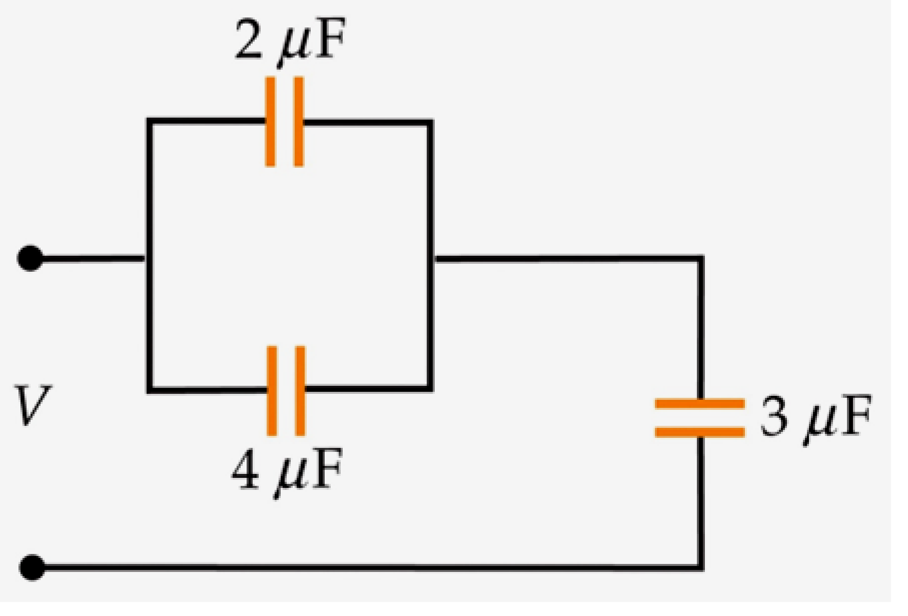


Equivalent capacitance C

Homework 9-8

(a) Find the equivalent capacitance of system of capacitors shown in the drawing

(b) Find the charge and potential difference (drop) on each of the capacitors assuming that the system has been powered by 6V-battery.



SUMMARY

- Electrostatics deals with the static electric fields generated by electric charges at rest.
- Electrostatic field is conservative. This field is characterized by both its electric field vector and scalar potential.
- Electric field can be determined based on the particular distribution of charge, using either Coulomb’s law and superposition principle or Gauss’ law.
- Capacitor is a device which stores the potential energy of the electric field. Energy density is proportional to E^2.
- Gauss’ law both in integral and differential forms represents one of the Maxwell equations.